A Semi-Automated Pipeline for Fast-Food Brand Exposure Detection in Kids' Screen-Recorded Media

KWOK Hoi Yi, Any

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Abstract

This study investigates the feasibility of a semi-automated pipeline for detecting brand exposure in screen-recorded data. The proposed framework integrates object detection, optical character recognition, and semantic reasoning to support large-scale analysis in scenarios where frame-level annotations are unavailable.

A staged evaluation strategy is adopted, comprising controlled experiments and large-scale application to real-world screen recordings collected as part of the Kids Online project. The results demonstrate that an ensemble combining YOLOv11 and YOLO-World provides robust performance under highly imbalanced and noisy conditions, enabling the identification of sparse and short-duration brand exposure events.

Rather than aiming to produce precise exposure measurements, the proposed system is designed as a scalable screening tool that reduces manual effort and supports downstream analysis. The findings suggest that integrating detection-based and semantic approaches offers a practical and extensible foundation for automated brand exposure analysis in real-world digital media environments.

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Chapter 1

Introduction

1.1 Problem Overview

The rapid expansion of digital technology has fundamentally reshaped how children access information and entertainment. With the widespread use of smartphones, tablets, and online video platforms, children now engage with digital media on a daily basis and often from a very young age. Compared with traditional television-based media, contemporary digital platforms provide on-demand access, personalised content, and continuous exposure to multimedia materials (Montgomery et al., 2019).

This transformation has also altered the way commercial content is presented. Advertising is no longer confined to clearly defined commercial breaks. Instead, brand-related content is increasingly embedded within entertainment material, appearing in the form of logos, product packaging, background visuals, or brief on-screen elements. Such exposure is often subtle and difficult to distinguish from non-commercial content, particularly for younger audiences (Russell, 2002; Smith et al., 2019; Jans et al., 2019; Kaye et al., 2020).

As a result, understanding the nature of children's exposure to branded content within digital environments has become an important research topic. However, the complexity and volume of modern digital media make systematic analysis increasingly challenging using traditional manual methods (Montgomery et al., 2019).

1.1.1 Kids Online Project

The Kids Online project was established to investigate children's real-world digital experiences through the collection of screen-recorded data. Unlike survey-based or self-reported approaches, the project captures actual on-screen activity, providing a direct and objective record of the content children encounter during everyday device use.

The raw data consists of long-duration screen recordings collected across a variety of contexts, including video viewing, application usage, and general device interaction. This form of data offers a unique opportunity to examine digital exposure in a naturalistic setting. At the same time, it introduces substantial analytical challenges due to its scale, diversity, and unstructured nature.

Manual inspection of such data is not feasible given the volume of recordings involved. Moreover, the content within screen recordings is highly dynamic, containing rapid scene changes, overlapping visual elements, and frequent transitions between applications. These characteristics necessitate the use of automated methods capable of processing visual data efficiently and consistently.

1.2 Research Focus and Objectives

Within the context of the Kids Online project, this study focuses on the semi-automated identification of brand exposure in screen-recorded media. The research aims to examine how existing computer vision and semantic understanding models can be effectively utilised and combined to support large-scale analysis of brand-related content in real-world digital environments (Montgomery et al., 2019).

However, translating this objective into practice introduces significant technical challenges, as screen-recorded media exhibit substantial variability in visual quality, layout, motion, and contextual structure across frames. Prior studies have noted that pixel-based detection models often struggle under such conditions, especially when logos are partially visible, visually ambiguous, or embedded within complex backgrounds (Chung et al., 2019; Wu et al., 2026).

To address these challenges, this study investigates an integrated detection framework that combines pixel-level object detection, optical character recognition (OCR), and semantic understanding models. Rather than treating these components in isolation, the research explores how their complementary strengths can be leveraged to improve robustness and reliability when analysing long-duration screen-recorded data, where single-model approaches are prone to instability and false positives (Yumang et al., 2022; V et al., 2024; Ren et al., 2024b; Zong et al., 2025).

The study adopts a system-oriented and empirical perspective, focusing on the practical behaviour of detection models under realistic conditions rather than idealised benchmark settings. In particular, it examines the effectiveness of combining YOLOv11 and YOLO-World within an ensemble framework, supported by textual cues extracted via optical character recognition (OCR) and semantic cues provided by vision–language models such as CLIP, to facilitate scalable brand exposure screening without reliance on dense frame-level ground truth annotations (Yin et al., 2016; Montgomery et al., 2019; Radford et al., 2021; Li et al., 2022; Zong et al., 2025).

The specific research objectives are as follows:

1. To examine the role of semantic representations, such as those derived from CLIP, in supporting pixel-level logo and brand detection within screen-recorded media.

2. To develop and evaluate a semi-automated brand exposure detection pipeline that integrates YOLOv11, YOLO-World, and auxiliary semantic cues for large-scale analysis.

3. To analyse the performance characteristics, strengths, and limitations of the proposed pipeline when applied to highly imbalanced, real-world screen-recorded data.

Based on these objectives, the following research questions are proposed:

1. To what extent can an ensemble-based pipeline combining YOLOv11 and YOLO-World support semi-automated brand exposure detection in screen-recorded data?

2. How does the proposed pipeline behave in large-scale real-world scenarios in terms of detection sparsity, stability, and false-positive tendencies?

The study is conducted using over 190 hours of screen-recorded video, from which more than 700,000 frames are extracted for analysis. This scale enables systematic evaluation of the proposed approach under realistic and highly unconstrained conditions representative of real-world digital media usage.

Chapter 2

Literature Review

This chapter reviews existing literature on logo detection and vision–language modelling and positions the present study within this intersection. The objective is to examine how semantic understanding models can be incorporated into a practical detection pipeline that produces interpretable and verifiable outputs, supporting large-scale analysis of brand appearance in screen-recorded data.

2.1 Challenges of Logo Detection

Logo detection is commonly formulated as a visual recognition task within the broader field of object detection. The primary objective is to identify and localise brand logos in images or video frames by learning discriminative visual patterns from labelled training data. With the advancement of deep learning techniques, convolutional neural networks have become the dominant approach for this task, enabling end-to-end learning of both visual features and spatial localisation and establishing object detection as a reliable foundation for logo recognition under controlled conditions (Benjelloun et al., 2020).

Early logo detection systems relied on hand-crafted features such as SIFT or HOG combined with traditional classifiers (Lowe, 2004; Dalal and Triggs, 2005). These approaches were later superseded by deep learning based detectors, particularly one-stage and two-stage object detection frame-works. Among these, single-stage detectors such as the YOLO family have achieved widespread adoption in applied visual recognition tasks due to their favourable balance between detection accuracy and computational efficiency(Benjelloun et al., 2020). By directly predicting bounding boxes and class probabilities from image features, such models have proven particularly effective for large-scale image and video analysis.

In controlled benchmark settings, logo detection models have demonstrated strong and reliable performance when trained on curated datasets containing clearly visible logos and limited background noise. Public benchmark datasets typically consist of centred, high-resolution logo instances with minimal occlusion, enabling models to learn discriminative visual cues effectively (Constantin and Ciocoiu, 2025; Jiang and Zhong, 2025). Under these conditions, pixel-based detection methods consistently achieve high accuracy and stable performance.

However, the assumptions underlying these datasets do not fully reflect real-world visual environments. In unconstrained screen-recorded videos, logos often appear at very small scales, are partially occluded, or remain visible only for brief durations. Additional challenges arise from motion blur, video compression artefacts, varying screen resolutions, and complex background clutter. Under such conditions, pixel-level detectors are prone to false positives and missed detections, particularly when background elements or packaging textures resemble brand logos (Chung et al., 2019; Wu et al., 2026).

2.2 Limitations of Pixel-based Logo Detection

Contemporary YOLO-based architectures, supported by large-scale pretraining, improved feature representations, and enhanced data augmentation strategies, are capable of handling many challenging conditions such as scale variation, partial occlusion, and background clutter. In practical applications, these models often achieve satisfactory performance across a wide range of visual scenarios (Constantin and Ciocoiu, 2025; Jiang and Zhong, 2025). However, these improvements do not fundamentally change the underlying nature of pixel-based detection.

First, conventional logo detectors rely heavily on bounding-box supervision. The detection process assumes that brand identity can be reliably inferred from a visually distinctive and spatially localised region, an assumption that often does not hold in real-world screen-recorded data (Chung et al., 2019; Wu et al., 2026).

Second, pixel-based detectors lack an explicit understanding of visual context. These models learn correlations between pixel patterns and class labels, but do not reason about the semantic meaning of the surrounding scene. Visually similar patterns may be incorrectly classified as brand logos, leading to false positive detections. For example, certain colour combinations, shapes, or textures may resemble a logo without actually representing a brand reference. This becomes particularly problematic in cases involving stylised logos, partial packaging, or scenes in which brand identity is implied rather than explicitly shown.

For these reasons, while object detection provides a strong and mature foundation for logo localisation, its inherent limitations motivate the exploration of complementary approaches. In particular, incorporating semantic-level understanding offers the potential to extend beyond appearance-based recognition and to capture higher-level brand-related cues that are difficult to model using pixel information alone.

2.3 Semantic Understanding via Vision–Language Models

Recent advances in vision–language models have introduced a new paradigm for visual understanding that extends beyond traditional pixel-based recognition. Large-scale vision–language pretraining has demonstrated strong capability in aligning visual content with semantic concepts, enabling more flexible and generalisable visual understanding across diverse tasks. Unlike conventional object detectors, which rely primarily on visual appearance and spatial features, vision–language models learn joint representations of images and natural language, enabling semantic-level interpretation of visual content.

A representative example of this paradigm is CLIP, which is trained on large-scale image–text pairs to align visual features with textual concepts in a shared embedding space. Rather than predicting a fixed set of object categories, CLIP enables similarity-based reasoning between images and language descriptions, allowing for flexible and open-vocabulary recognition.(Radford et al., 2021).

The semantic nature of vision–language models allows them to capture higher-level information that pixel-based detectors cannot easily represent. For example, even when a brand logo is partially visible or absent, a model may infer brand-related context based on food presentation, packaging style, or scene composition. Such semantic reasoning is especially relevant in real-world screen-recorded data, where visual conditions are uncontrolled and brand appearances are often subtle or indirect.

Building on this paradigm, recent research has explored how semantic understanding can be incorporated into object detection frameworks, rather than being used solely for global image-level reasoning. Approaches such as Grounding DINO and OWL-ViT incorporate language conditioning into detection frameworks, enabling open-vocabulary localisation based on textual prompts (Ren et al., 2024a; Chhipa et al.,

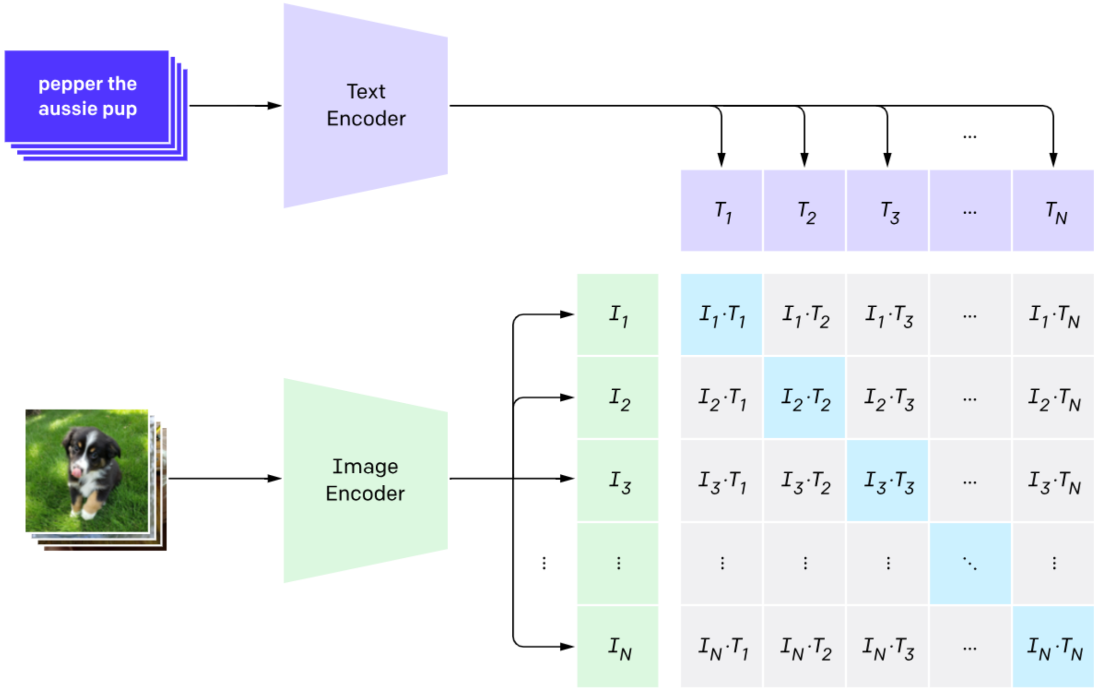


Figure 2.1: CLIP architecture overview illustrating joint image–text embedding and contrastive similarity learning. Source: CLIP documentation Radford et al. (2021).

2024). While these models demonstrate strong semantic grounding capability, they rely heavily on transformer-based architectures and cross-modal attention mechanisms. As a result, their computational cost is high, and practical deployment typically requires GPU acceleration, making them less suitable for large-scale or long-duration video analysis.

In contrast, YOLO-World represents a more practical adaptation of the vision–language paradigm. By integrating language embeddings into a YOLO-style detection framework, YOLO-World enables semantic-guided object detection while preserving the computational efficiency and architectural simplicity of the YOLO family. Unlike pure vision–language models, YOLO-World does not perform full cross-modal reasoning; instead, textual information serves as a conditioning signal that guides detection. This design allows the model to maintain efficient inference while still benefiting from semantic awareness.

By default, YOLO-World is pretrained on large-scale datasets COCO50, which allows a degree of zero-shot generalisation to generic object categories. However, for more specific tasks such as brand or logo detection, further refinement is typically required. This refinement may reduce some open-vocabulary flexibility, but it provides a reasonable and necessary trade-off by improving detection stability and task-specific reliability (Cheng et al., 2024).

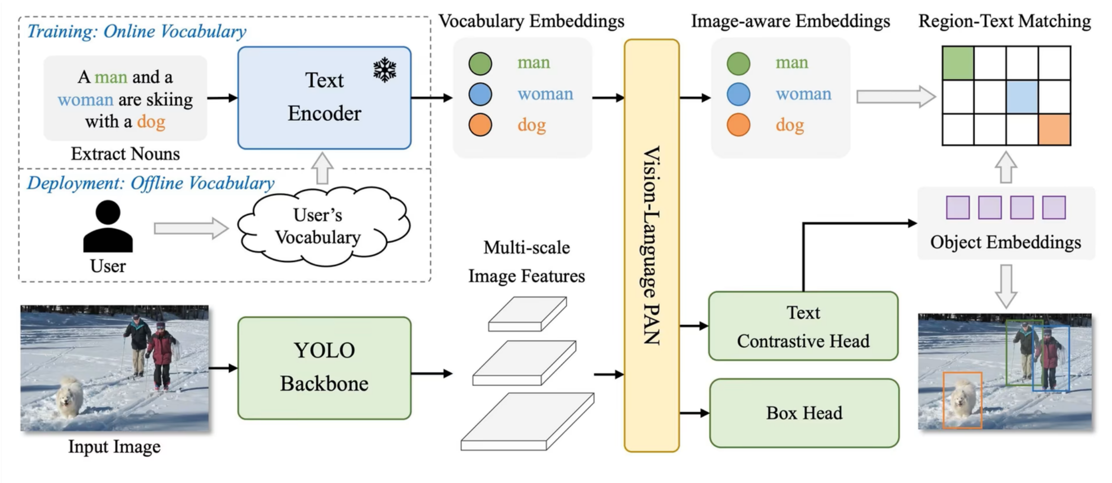


Figure 2.2: YOLO-World architecture illustrating semantic-guided object detection with vision–language conditioning. Source: Ultralytics documentation (Cheng et al., 2024).

Although a direct benchmark comparison is not the focus of this study, the architectural differences between transformer-based detectors and YOLO-style detectors are well established. The latter are widely adopted in real-time and large-scale applications due to their substantially lower computational cost. YOLO-World follows this design philosophy, making it feasible for deployment under relatively modest hardware constraints.

2.4 Complementary Role of Semantic Models in Logo Detection

While both pixel-based object detectors and vision–language models demonstrate strong capabilities in their respective domains, neither approach alone is sufficient for robust logo detection in real-world screen-recorded data. Recent studies have shown that combining these approaches can yield more robust visual understanding under challenging conditions. Pixel-based detectors excel at spatial localisation but lack semantic understanding, whereas semantic models offer conceptual reasoning but are limited in localisation accuracy and computational efficiency. This complementary relationship motivates the adoption of a hybrid framework that integrates both paradigms (Chung et al., 2019; Radford et al., 2021; Ren et al., 2024b).

Under real-world screen-recorded conditions, pixel-based detection alone often struggles due to weak or ambiguous visual cues, while semantic models used in isolation lack reliable spatial localisation and may produce overly general predictions (Chhipa et al., 2024; Constantin and Ciocoiu, 2025; Wu et al., 2026).

Semantic models contribute value by introducing contextual understanding. Through vision–language alignment, these models can associate visual content with higher-level concepts such as brand-related styles, packaging types, or consumption contexts. This allows them to capture information that is not explicitly encoded in pixel patterns (Radford et al., 2021; Zong et al., 2025). For example, even when a logo is not fully visible, semantic cues related to food presentation or brand aesthetics may still indicate a likely brand presence.

Furthermore, brand-related visual patterns are subject to temporal variation, introducing distribution shifts that are difficult to capture using static training data alone. While this study does not explicitly model temporal trends, semantic representations provide a degree of flexibility by relying on conceptual similarity rather than fixed visual templates (Jiang and Zhong, 2025).

However, semantic models also introduce new limitations. A key challenge lies in the interpretability of text embeddings. Unlike bounding boxes or pixel-level features, semantic representations function largely as black-box abstractions, making it difficult to determine which visual elements contribute to a particular prediction. This reduces transparency and complicates error analysis, particularly when false positives occur (Chhipa et al., 2024).

Within this study, semantic understanding is therefore incorporated as an auxiliary component rather than a replacement for pixel-based detection. Models such as CLIP and YOLO-World are used to provide semantic guidance and contextual validation, while conventional detectors remain responsible for precise localisation. Importantly, YOLO-World is treated not as a full vision–language model, but as a semantic-guided detector. By integrating language embeddings into a YOLO-style architecture, it preserves the efficiency and deployment advantages of conventional detectors while benefiting from semantic conditioning. This design avoids the heavy computational overhead associated with transformer-based models such as Grounding DINO or OWL-ViT, making it more suitable for large-scale or long-duration video analysis (Cheng et al., 2024).

Overall, the hybrid approach adopted in this study reflects a pragmatic balance between semantic understanding and computational feasibility. By combining pixel-level localisation with semantic guidance, the proposed framework aims to improve robustness, interpretability, and scalability in real-world brand detection tasks, forming a solid foundation for the methodology presented in the following chapter.

Chapter 3

Methods and Validation

This chapter presents the methodology adopted in this study for developing and evaluating a brand exposure detection framework based on computer vision techniques. The overall design follows a staged evaluation strategy, progressing from controlled model comparison to large-scale robustness cross-checking under realistic deployment conditions. This design is intentionally adopted to ensure that model selection is not driven solely by benchmark performance, but is also informed by practical constraints associated with real-world exposure analysis, including computational efficiency, robustness under class imbalance, and false positive control (Chung et al., 2019; Hawke, 2025).

A key methodological decision in this study is to treat brand logos as the primary visual evidence for brand identification. This choice is motivated by both feasibility and consistency. Food imagery alone is often ambiguous in terms of brand attribution, while product packaging can vary significantly across product lines, geographic regions, and time periods, making it difficult to define stable visual categories. In contrast, brand logos provide a relatively consistent and recognisable visual cue that can be operationalised more reliably across diverse real-world screen recordings.

Five major fast-food brands: Burger King, KFC, McDonald's, Domino's Pizza, and Pizza Hut were selected due to their widespread market presence in New Zealand, and their relevance to public health research on food marketing exposure (Smith et al., 2019; Hawke, 2025). All experiments, evaluations, and exposure analyses in this chapter are restricted to this fixed set of target brands to ensure consistency across experimental stages and to support interpretable exposure measurement.

Model selection rationale Two object detection models, YOLOv11 and YOLO-World, are selected as the core detection components of the proposed framework base on a cost–performance trade-off.

YOLOv11 was selected as one of the primary fully supervised detectors due to prior empirical evidence indicating its strong performance on video-based and screen-rendered visual data (Jiang and Zhong, 2025). YOLO-World is incorporated as a complementary detector, allowing language-aligned information to be introduced while preserving the computational efficiency of the YOLO architecture (Cheng et al., 2024; Jiang and Zhong, 2025). Importantly, YOLO-World is not treated as a full grounding-based model in this study, but rather as a semantic-augmented extension of YOLO. This distinction allows semantic information to be incorporated without introducing the substantial overhead and confidence instability commonly associated with heterogeneous grounding pipelines (Singh et al., 2025; Jiang and Zhong, 2025).

The exclusive use of YOLO-family models further ensures architectural consistency across detectors, enabling meaningful comparison and stable ensemble integration. Prior work has shown that combining detectors with incompatible confidence semantics can lead to unstable behaviour under severe class imbalance, particularly in exposure-oriented applications where false positives carry disproportionate cost (Vargoorani et al., 2025; Singh et al., 2025).

Staged evaluation strategy The methodological framework consists of two main stages. The first stage focuses on model development and baseline evaluation under controlled conditions using instance-level metrics. This stage is designed to identify stable and competitive model configurations while minimising confounding factors introduced by deployment noise. The second stage extends the evaluation from controlled model comparison to exposure-oriented analysis. While instance-level evaluation is suitable for measuring object detection performance, it does not adequately reflect the requirements of exposure analysis, where the primary concern is whether a brand appears within a video frame rather than how accurately individual objects are localised (Russell, 2002; Hawke, 2025).

To address this gap, the evaluation framework transitions from instance-level detection to frame-level multi-class evaluation, in which each frame is treated as a single prediction unit. Model selection and parameter tuning are therefore conducted in a staged manner: initial filtering is performed using instance-level metrics to identify candidate models, after which frame-level evaluation is applied to assess performance under more realistic conditions. Finally, model robustness is examined using large-scale raw video data to evaluate false positive behaviour under extreme class imbalance, a condition commonly encountered in real-world exposure detection tasks (Chhipa et al., 2024; Wu et al., 2026).

This progressive evaluation strategy ensures that the final model configuration is not only accurate under controlled settings, but also reliable when deployed on real-world video data. Together, these stages form a coherent methodological pipeline that links controlled model evaluation with practical exposure analysis. The following sections describe each stage in detail, beginning with the construction of training datasets and baseline evaluation procedures in the next section.

3.1 Model development and baseline evaluation

3.1.1 Training Datasets and Built-in Evaluation

To support a systematic comparison of object detection performance under progressively realistic conditions, three training datasets with increasing levels of visual complexity were constructed using Roboflow (Roboflow, 2025), as summarised in Table 3.1. These datasets differ in data source, visual characteristics, and noise level, ranging from clean photographic images to frames extracted directly from screen-recorded videos. This staged dataset design follows common practice in object detection research, where model behaviour is examined across varying levels of visual difficulty (Benjelloun et al., 2020; Jiang and Zhong, 2025).

CleanLogo-Only The CleanLogo-Only dataset consists of clean logo and product images collected from photographic sources rather than video recordings. These images typically exhibit clear object boundaries, stable lighting conditions, and minimal background clutter. While still representative of real-world brand appearances, they lack the motion blur, compression artefacts, and visual noise commonly observed in screen-recorded footage. As such, this dataset serves as an idealised training condition and provides an upper-bound reference for model performance rather than a realistic deployment scenario (Constantin and Ciocoiu, 2025).

CleanLogo + VideoMix The CleanLogo + VideoMix dataset extends the first by incorporating frames extracted directly from screen-recorded videos, together with an increased proportion of dummy negative samples. Compared with photographic images, these video-derived frames introduce additional challenges, including motion blur, compression artefacts, partial occlusion, scale variation, and complex background content. Consequently, this dataset more closely reflects the visual characteristics of real-world Kids Online recordings, where visual conditions are uncontrolled and highly variable (Chung et al., 2019; Hawke, 2025).

CleanLogo + VideoMix + HardNeg The CleanLogo + VideoMix + HardNeg dataset further increases realism by augmenting the second dataset with hard negative samples collected from previous pilot experiments. These samples are intentionally selected to resemble visually ambiguous non-brand content that frequently leads to false detections, such as food-like objects, packaging with similar colour schemes, or interface elements. This dataset is specifically designed to stress-test model precision and false positive behaviour, which is critical for exposure analysis where misclassification can lead to inflated exposure estimates.

Table 3.1: Summary of training datasets with increasing visual complexity. The three datasets progress from clean photographic images to video-derived frames and hard negative samples, reflecting progressively more realistic deployment conditions.

Dataset	Brand	Clean Logo	Product	Video	Total
CleanLogo-Only	Burger King	202	118	–	
	Domino's	193	93	–	
	KFC	197	104	–	
	McDonald's	216	102	–	
	Pizza Hut	187	74	–	
	Dummy (∼5%)	70	–	–	**1556**
CleanLogo + VideoMix	Burger King	202	118	62	
	Domino's	193	93	76	
	KFC	98	104	72	
	McDonald's	217	102	64	
	Pizza Hut	187	74	93	
	Dummy (∼15%)	275	–	–	**2030**
CleanLogo + VideoMix + HardNeg	Burger King	202	118	62	
	Domino's	193	93	76	
	KFC	98	104	72	
	McDonald's	217	102	64	
	Pizza Hut	187	74	93	
	Dummy (∼15%)	270	–	–	**2025**

For each dataset, three training scales are defined: small (S), medium (M), and large (L). The small-scale dataset uses only the original images without multiplicative augmentation. The medium-scale dataset approximately doubles the number of training samples through data augmentation, while the large-scale dataset further increases the dataset size to approximately three times that of the base set. This scaling strategy is designed to examine whether increasing data volume through conventional pixel-level augmentation leads to measurable performance gains when training modern YOLO-family detectors.

Data augmentation operations include horizontal flipping, $\pm 5^\circ$ rotation, $\pm 20\%$ brightness adjustment, $\pm 10\%$ exposure variation, and mild blurring with a kernel size of 0.5 pixels. Such augmentation strategies are widely used in object detection to improve generalisation under visual variability without altering semantic identity (Benjelloun et al., 2020; Constantin and Ciocoiu, 2025). These transformations are applied consistently across datasets to simulate realistic visual variation while preserving brand identity.

Table 3.2: Composition of training, validation, and test splits under different dataset scales (S, M, L). Larger scales are obtained through multiplicative data augmentation applied to the base dataset.

Dataset	Size	Train	Val	Test	Total
CleanLogo-Only	S	1072	299	148	1519
	M	1720	440	219	2379
	L	2217	519	261	2997
CleanLogo + VideoMix	S	1496	393	193	2082
	M	2384	596	294	3274
	L	2691	789	396	3876
CleanLogo + VideoMix + HardNeg	S	1477	400	200	2077
	M	2090	598	299	2987
	L	2450	720	360	3530

All models are trained and evaluated using the built-in instance-level evaluation procedures provided by the YOLO framework. This ensures that performance comparisons between YOLOv11 and YOLO-World are conducted under a consistent training and evaluation protocol, thereby minimising confounding effects introduced by heterogeneous optimisation or metric implementations.

Performance is assessed using standard object detection metrics, primarily F1 score and mAP@0.5, with recall examined only as a supplementary indicator to ensure that detection sensitivity is not excessively degraded. This stage provides an initial performance characterisation of all trained models before transitioning to frame-level, exposure-oriented evaluation in the subsequent stage (Table 3.3 and Table 3.4).

Table 3.3: YOLOv11 Performance Across Training Datasets

Dataset	Size	mAP_{50}	F1	Recall
CleanLogo-Only	S	0.929	0.90	0.96
	M	0.912	0.89	0.94
	L	0.909	0.88	0.93
CleanLogo + VideoMix	S	0.938	0.90	0.96
	M	0.917	0.88	0.94
	L	0.919	0.89	0.93
CleanLogo + VideoMix + HardNeg	S	0.936	0.89	0.96
	M	0.933	0.90	0.95
	L	0.911	0.88	0.92

Table 3.4: YOLO-World Performance Across Training Datasets

Dataset	Size	mAP_{50}	F1	Recall
CleanLogo-Only	S	0.935	0.90	0.96
	M	0.838	0.88	0.93
	L	0.913	0.88	0.93
CleanLogo + VideoMix	S	0.940	0.90	0.97
	M	0.916	0.89	0.93
	L	0.910	0.89	0.92
CleanLogo + VideoMix + HardNeg	S	0.945	0.90	0.97
	M	0.924	0.90	0.94
	L	0.917	0.89	0.93

3.1.2 Comparative Observations from Built-in Evaluation

Across all datasets and training scales, YOLOv11 and YOLO-World demonstrate broadly comparable performance, with instance-level F1 scores consistently ranging between approximately 0.88 and 0.90. No single model exhibits a clear and consistent advantage across all experimental settings. However, systematic differences in model behaviour can be observed, reflecting trade-offs commonly reported in comparative studies of object detection models.

Under built-in instance-level evaluation, YOLOv11 and YOLO-World exhibit highly similar recall performance across all datasets and training scales, with recall values generally exceeding 0.92 under default confidence settings. No consistent recall advantage is observed for either model at this stage. Despite similar aggregate metrics, the two models differ in their detection behaviour. YOLOv11 tends to produce more confident and concentrated predictions, whereas YOLO-World incorporates language-aligned representations that constrain detections semantically. Similar behavioural differences between conventional YOLO-style detectors and semantic-guided models have been reported in prior comparative studies (Cheng et al., 2024; Chhipa et al., 2024). These differences are not strongly reflected in instance-level metrics, but become more apparent under deployment-oriented evaluation settings, as examined in later stages of this study.

Increasing training scale through data augmentation does not lead to substantial performance gains. In most cases, models trained on the base-scale dataset achieve performance comparable to, or marginally better than, those trained on medium- or large-scale variants. The only notable improvement is observed in the most noise-intensive dataset, where the medium-scale YOLOv11 model exhibits a modest increase in F1 score. Overall, these results indicate that conventional pixel-level augmentation contributes limited additional discriminative power, particularly when models are already initialised from strong pretrained representations. This pattern aligns with previous findings that conventional pixel-level augmentation often yields diminishing returns under strong pretraining regimes (Benjelloun et al., 2020; Jiang and Zhong, 2025).

Taken together, these findings suggest that dataset realism and negative sample composition exert a greater influence on model behaviour than training scale alone. While instance-level evaluation is effective for controlled model comparison and initial model filtering, it does not fully capture performance under real-world deployment conditions. In particular, when applied to large-scale raw video data without curated annotations, models may exhibit increased sensitivity to unfamiliar background patterns, leading to elevated false positive rates. Similar robustness challenges under distribution shift have been reported in recent studies of object detection models (Chhipa et al., 2024; Wu et al., 2026).

These observations highlight the limitations of relying solely on instance-level metrics for exposure analysis and motivate the need for additional evaluation stages and filtering mechanisms. Accordingly, the next sections extend the evaluation from instance-level detection to frame-level, exposure-oriented validation, in line with methodological recommendations for applying detection models to highly imbalanced, real-world visual data (Hawke, 2025).

3.2 Transition from Instance-based to Frame-level Multi-class Evaluation

While instance-based evaluation is effective for comparing object detection models under controlled conditions, it is insufficient for the purpose of brand exposure analysis. Instance-level metrics primarily assess bounding box localisation accuracy and object classification performance, whereas the central objective of this study is to determine whether a brand appears within a video frame. From an exposure analysis perspective, the presence of a brand within a frame is of greater importance than the number of detected instances or their spatial precision (Russell, 2002; Hawke, 2025).

To better align the evaluation framework with the objective of exposure analysis, the evaluation is extended from instance-based detection to frame-level multi-class evaluation, a formulation that has been adopted in prior studies examining object presence in video-based and exposure-oriented analysis (Chung et al., 2019). Each frame is treated as a single prediction unit. A frame is considered positive for a given brand if that brand appears at least once within the frame, regardless of the number of detected instances or their spatial locations.

When multiple detections of the same brand occur within a single frame, the highest confidence score is used to represent the frame-level prediction. This design prevents repeated detections of the same object from artificially inflating evaluation results and ensures a consistent and conservative interpretation of brand exposure, in line with prior frame-level and video-level detection practices (Gautam et al., 2023).

To support this transition, two frame-base validation datasets are constructed using images extracted directly from screen-recorded videos, ensuring consistency with the data used in the subsequent exposure analysis. The first dataset, the instance-level validation set, consists primarily of frames containing at least one target brand and is used to evaluate model behaviour under noisy, real-world visual conditions. This dataset contains 503 frames, with approximately 100 images per brand. A small number of frames without brand annotations are included due to annotation noise; however, their proportion is negligible and does not materially affect the evaluation.

Building upon this dataset, a second validation dataset, the frame-level validation set, is constructed by incorporating 487 dummy negative frames containing no brand exposure, together with 71 additional positive frames collected from previous studies. The resulting dataset is approximately balanced, with a near 1:1 ratio between positive and negative samples. This design is intended to prevent evaluation metrics from being dominated by negative samples and to enable stable threshold selection under conditions that more closely resemble real-world video data, where exposure events are sparse but non-negligible.

Under this frame-level evaluation setting, performance is assessed using frame-based true positives, false positives, and false negatives. A true positive corresponds to a frame in which a target brand is present and correctly detected, while a false positive refers to a frame without brand exposure that is incorrectly classified as containing a brand. This formulation more accurately reflects the requirements of exposure analysis, where the key concern is whether exposure occurs rather than the precise localisation of individual objects.

Using this frame-level validation framework, candidate models selected from the instance-level evaluation stage are further compared. Model selection at this stage is primarily based on frame-level F1 score, which balances recall and precision under exposure-oriented evaluation. The selected model configurations are then carried forward for pipeline construction and large-scale validation on raw video data.

In addition to the two validation datasets, a large-scale real-world dataset consisting of approximately 55,000 frames is used for final stress testing. This dataset contains only 71 confirmed brand-positive frames and therefore exhibits extreme class imbalance. It is not used for model selection or threshold tuning, but solely to evaluate robustness and false positive behaviour under realistic deployment conditions, where even a very low false positive rate can substantially affect exposure estimates (Hawke, 2025; Wu et al., 2026).

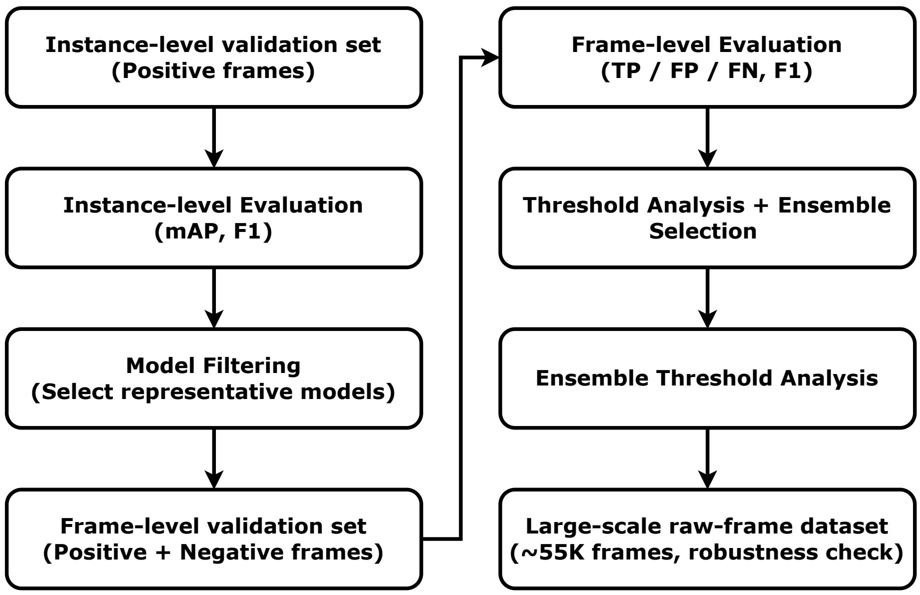


Figure 3.1: Evaluation pipeline illustrating the transition from instance-level model comparison to frame-level, exposure-oriented validation.

3.2.1 Evaluation Setup and Model Selection

To enable the transition toward frame-level multi-class detection, all eighteen trained models are first evaluated using the instance-level validation dataset under the built-in instance-based evaluation protocol. At this stage, instance-level evaluation is used solely as a filtering mechanism to identify stable and competitive model configurations, rather than as a final indicator of deployment suitability. From each dataset and model family, only the best-performing model is retained, resulting in a total of six candidate models for subsequent analysis. This filtering step reduces redundancy and ensures that later evaluations focus on representative and competitive model configurations. The performance of the models are summarised in Table 3.5 and Table 3.6.

Table 3.5: YOLOv11 instance-level evaluation results.

Model	mAP_{50}	F1
CleanLogo-Only-S	0.7636	0.7459
CleanLogo-Only-M	0.7819	0.7449
CleanLogo-Only-L	**0.7907**	0.7463
CleanLogo + VideoMix-S	0.8260	0.7991
CleanLogo + VideoMix-M	**0.8345**	0.8004
CleanLogo + VideoMix-L	0.7883	0.7641
CleanLogo + VideoMix + HardNeg-S	**0.8212**	0.7843
CleanLogo + VideoMix + HardNeg-M	0.8171	0.7766
CleanLogo + VideoMix + HardNeg-L	0.8072	0.7803

Table 3.6: YOLO-World instance-level evaluation results.

Model	mAP_{50}	F1
CleanLogo-Only-S	0.7905	0.7564
CleanLogo-Only-M	**0.7951**	0.7558
CleanLogo-Only-L	0.7794	0.7720
CleanLogo + VideoMix-S	0.8290	0.8024
CleanLogo + VideoMix-M	**0.8412**	0.8173
CleanLogo + VideoMix-L	0.8178	0.7883
CleanLogo + VideoMix + HardNeg-S	**0.8207**	0.7891
CleanLogo + VideoMix + HardNeg-M	0.8072	0.7719
CleanLogo + VideoMix + HardNeg-L	0.7994	0.7644

Compared with the training-stage evaluation, a different performance pattern emerges. While most models perform well on curated training data, those trained primarily on cleaner datasets exhibit reduced robustness when applied to video-extracted frames, reflecting a common generalisation challenge under distribution shift (Chung et al., 2019; Chhipa et al., 2024).

In particular, models trained on the CleanLogo-Only and CleanLogo + VideoMix dataset groups benefit from additional data augmentation when evaluated under frame-base conditions. Augmentations such as horizontal flipping, mild rotation, brightness or exposure adjustment, and blurring are observed to improve robustness. This observation is consistent with prior findings that augmentation can partially compensate for limited visual diversity in training data, especially when models are deployed in noisier real-world video environments (Constantin and Ciocoiu, 2025; Wu et al., 2026). These results indicate that models trained on visually constrained data distributions do not generalise optimally to noisier real-world video content and therefore require either augmentation or confidence calibration when applied to frame-level exposure analysis.

3.3 Frame-level Multi-class Evaluation Results

Following the instance-level model selection described in the previous section, the six selected models are further evaluated under a frame-level, multi-class setting using the frame-level validation set. This dataset contains both positive and negative samples in an approximately 1:1 ratio and is designed to better approximate real-world video conditions, where both exposure and non-exposure frames are present.

For each selected model, a confidence threshold sweep is performed over the range of 0.30 to 0.85 to analyse performance sensitivity under different decision thresholds. Figure 3.2 illustrates the frame-level F1 performance of the selected models across different confidence thresholds. The results indicate that optimal performance is consistently achieved when the confidence threshold lies between 0.30 and 0.50.

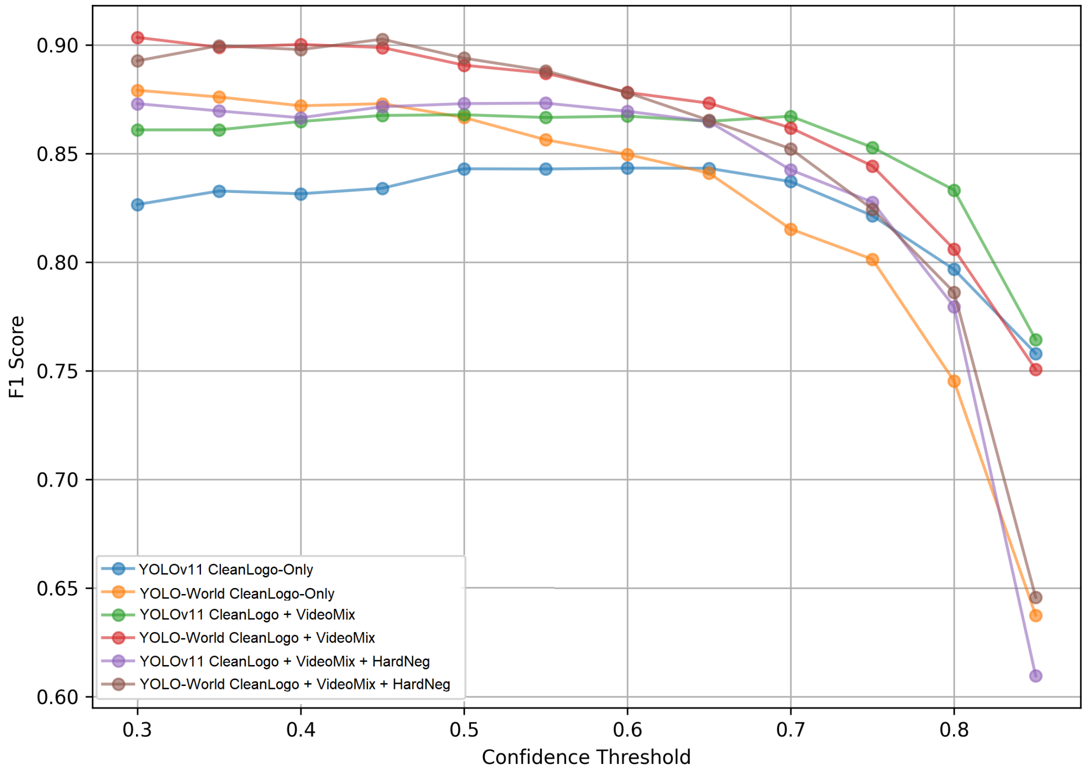


Figure 3.2: Frame-level performance under different confidence thresholds

As the confidence threshold increases, the false positive rate decreases; however, this improvement is accompanied by a noticeable reduction in recall, leading to a decline in overall F1 score beyond a certain point. This behaviour reflects the well-established trade-off between detection sensitivity and false positive control, which is particularly critical in exposure-oriented evaluation settings where the costs of missed detections and spurious detections are asymmetric.

Furthermore, distinct behavioural differences are observed between model families. Under the frame-level evaluation setting, YOLO-World models tend to achieve higher recall at lower confidence thresholds, while maintaining competitive false positive rates. In contrast, YOLOv11 models exhibit more conservative behaviour, achieving higher precision at the cost of recall. Similar precision–recall asymmetries between semantic-guided detectors and conventional YOLO-style models have been reported in recent comparative studies (Cheng et al., 2024; Chhipa et al., 2024). These complementary characteristics motivate the use of ensemble strategies to balance detection performance and robustness.

Based on these results, the best-performing YOLOv11 model (CleanLogo + VideoMix + HardNeg -S) and YOLO-World model (CleanLogo + VideoMix -M) are selected according to their frame-level F1 performance on the frame-level validation set. The detailed performance of the top-performing configurations is summarised in Table 3.7 (YOLOv11) and Table 3.8 (YOLO-World), which serve as the basis for subsequent ensemble construction.

Table 3.7: Top–5 YOLOv11 configurations under frame-level evaluation, ranked by F1 score.

Model	Confidence level	Precision	Recall	F1 score	FP rate
CleanLogo + VideoMix + HardNeg-S	0.55	0.9631	0.7986	**0.8732**	0.0160
CleanLogo + VideoMix + HardNeg-S	0.50	0.9474	0.8094	0.8729	0.0236
CleanLogo + VideoMix + HardNeg-S	0.30	0.9086	0.8399	0.8729	0.0443
CleanLogo + VideoMix + HardNeg-S	0.45	0.9415	0.8112	0.8715	0.0264
CleanLogo + VideoMix + HardNeg-S	0.35	0.9163	0.8273	0.8696	0.0396

3.4 Ensemble Model Selection and Trade-off Analysis

Following the frame-level evaluation described in the previous section, an ensemble strategy is constructed using the best-performing YOLOv11 and YOLO-World models. Given the highly imbalanced nature of real-world video data, in which negative

Table 3.8: Top–5 YOLO-WORLD configurations under frame-level evaluation, ranked by F1 score.

Model	Confidence level	Precision	Recall	F1 score	FP rate
CleanLogo + VideoMix-M	0.30	0.9540	0.8579	**0.9034**	0.0217
CleanLogo + VideoMix + HardNeg-S	0.45	0.9730	0.8417	0.9026	0.0123
CleanLogo + VideoMix-M	0.40	0.9650	0.8435	0.9002	0.0160
CleanLogo + VideoMix + HardNeg-S	0.35	0.9500	0.8543	0.8996	0.0236
CleanLogo + VideoMix-M	0.35	0.9573	0.8471	0.8989	0.0198

frames overwhelmingly outnumber positive ones, controlling false positives is a critical requirement for practical deployment (Russell, 2002; Hawke, 2025).

To address this, ensemble weights and confidence thresholds are systematically explored to analyse the trade-off between detection performance and false positive behaviour. Multiple ensemble configurations are evaluated under fixed false positive rate constraints of 0.5%, 0.4%, 0.3%, and 0.2%, in addition to the configuration achieving the highest overall F1 score.

The analysis indicates that while the ensemble configuration achieving the highest F1 score provides marginally improved detection performance, it is also associated with a higher false positive rate. In contrast, several alternative configurations achieve comparable frame-level performance while substantially reducing false positives. This observation suggests that maximising F1 score alone is insufficient for selecting an operational model in real-world deployment scenarios, particularly for exposure analysis where false alarms can inflate exposure estimates (Russell, 2002).

Figures 3.3-3.5 jointly illustrate the trade-off between detection performance and false positive behaviour under different ensemble configurations. Across a wide range of confidence thresholds and model weight combinations, a stable performance plateau is observed when the confidence threshold lies between approximately 0.30 and 0.50 and the weight assigned to YOLOv11 ranges from 0.3 to 0.6. Within this region, frame-level F1 performance remains relatively stable, while false positive rates decrease monotonically as the confidence threshold increases.

These results indicate that configurations achieving the highest F1 score are often associated with elevated false positive rates, whereas several alternative ensemble settings achieve comparable detection performance with substantially improved false positive control.

Table 3.9 summarises representative ensemble configurations under different false positive rate constraints and supports the selection of the final operating point.

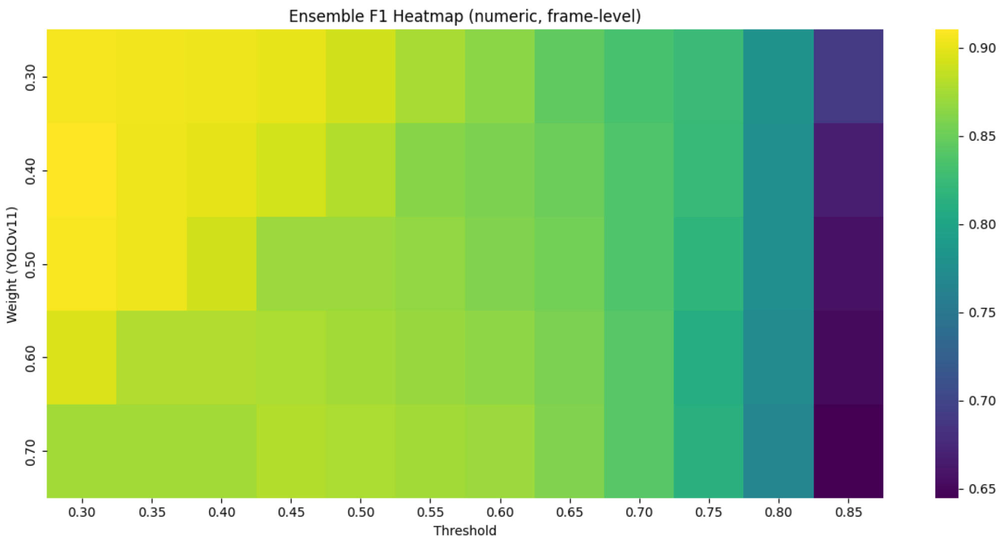


Figure 3.3: Frame-level F1 performance of the ensemble under different combinations of model weights and confidence thresholds.

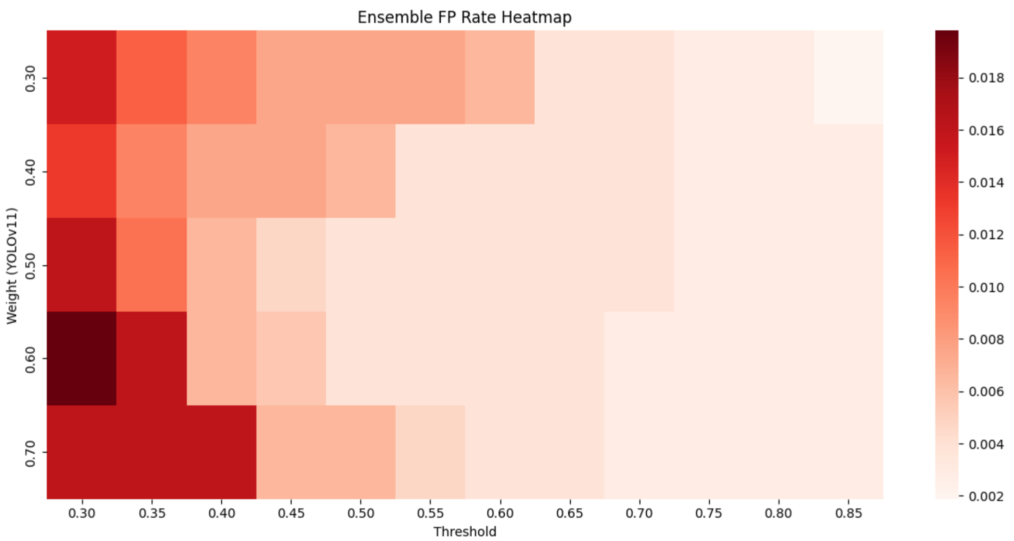


Figure 3.4: False positive rates of the ensemble under different combinations of model weights and confidence thresholds.

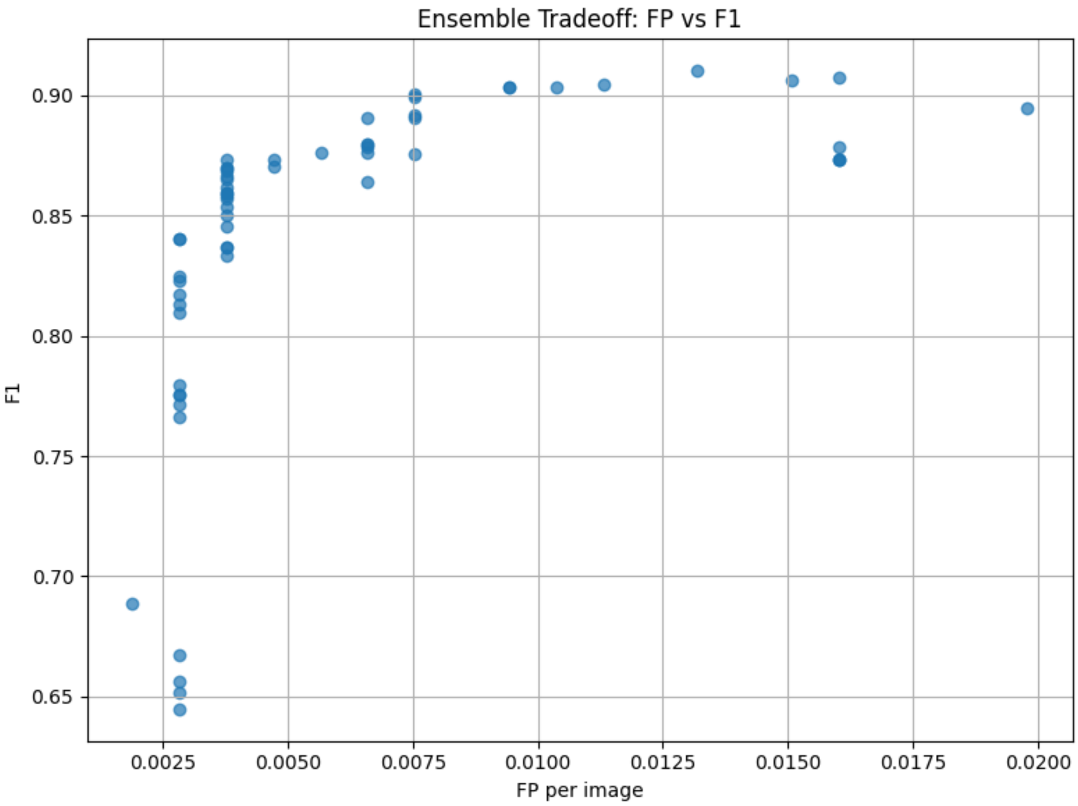


Figure 3.5: Trade-off between frame-level F1 score and false positive rate under different ensemble configurations.

Based on this analysis, a more conservative ensemble configuration is selected. Specifically, the ensemble with a YOLOv11:YOLO-World weight ratio of 0.6:0.4 and a confidence threshold of 0.50 is adopted, as it achieves a favourable balance between detection performance (F1 = 0.873) and false positive rate (0.38%), providing a more reliable operating point for large-scale video analysis.

Table 3.9: Summary of ensemble model selection on the frame-level validation set.

Model	Confidence level	F1 score	FP rate	Remarks
Ensemble (v11:wd = 4:6)	0.30	0.9101	0.0132	Overall best F1
YOLO-WORLD - CleanLogo + VideoMix-M	0.30	0.9034	0.0217	Base YOLO-WORLD
Ensemble (v11:wd = 7:3)	0.55	0.8735	0.0047	FP 0.5% best F1
Ensemble (v11:wd = 6:4)	**0.50**	**0.8732**	**0.0038**	**FP 0.4% best F1**
YOLOv11 - CleanLogo + VideoMix + HardNeg-S	0.55	0.8732	0.0160	Base YOLOv11
Ensemble (v11:wd = 6:4)	0.70	0.8402	0.0028	FP 0.3% best F1
Ensemble (v11:wd = 3:7)	0.85	0.6886	0.0019	FP 0.2% best F1

3.5 Large-scale Cross-check on Raw Frames and Ensemble Threshold Sweeping

While frame-level evaluation on the frame-level validation set provides a controlled basis for model and ensemble selection, it does not fully reflect real-world deployment conditions. In practice, brand exposure events are extremely sparse relative to the total volume of recorded content. To assess the robustness of the selected ensemble under realistic conditions, a large-scale cross-check is therefore conducted using the large-scale raw frame dataset.

The large-scale raw frame dataset consists of 54,840 frames extracted from approximately 15 hours of screen recordings. Brand occurrences within this dataset are highly imbalanced: Burger King appears in 4 frames, KFC in 22 frames, McDonald's in 40 frames, Pizza Hut in 5 frames, and no Domino's Pizza instances are observed. This distribution closely reflects real-world exposure scenarios, where the vast majority of frames contain no brand-related content.

Such extreme class imbalance poses a significant challenge for exposure detection, as even a very small false positive rate can lead to a large number of erroneous detections when scaled to long-duration recordings. Consequently, evaluation on raw data is essential to verify that the selected ensemble remains reliable beyond controlled validation settings (Chhipa et al., 2024; Wu et al., 2026).

In addition to overall performance verification, a confidence threshold sweep is performed on the selected ensemble model using the raw dataset. The purpose of this sweep is not to optimise the F1 score, but to examine whether the threshold selected from the frame-level validation set lies within a stable operating region, one in which false positives are sufficiently suppressed while the ability to detect rare brand exposures is preserved.

Given the extreme class imbalance, conventional balanced metrics such as F1 score become less informative in this setting. Greater emphasis is therefore placed on false positives per frame and qualitative inspection of detection results. Similar evaluation adaptations have been advocated in applied detection and exposure studies, where the cost of false positives outweighs marginal gains in recall (Russell, 2002; Hawke, 2025).

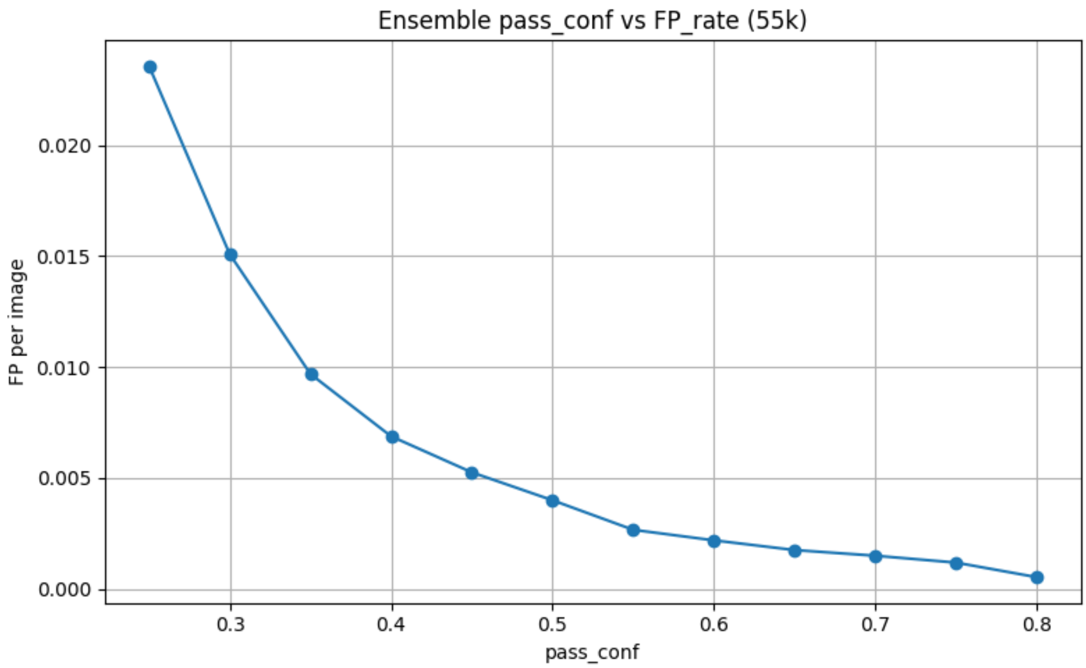


Figure 3.6: False positives per frame as a function of ensemble confidence threshold.

Figure 3.6 illustrates the relationship between the ensemble decision threshold ($pass\_conf$) and false positives per frame on the large-scale raw frame dataset, while Figure 3.7 presents the corresponding recall. As the threshold increases, false positives decrease monotonically, whereas recall gradually declines, demonstrating a clear trade-off between false positive suppression and detection sensitivity.

The observed trends confirm that the threshold selected from the validation stage remains well-behaved under real-world conditions, supporting its use for subsequent

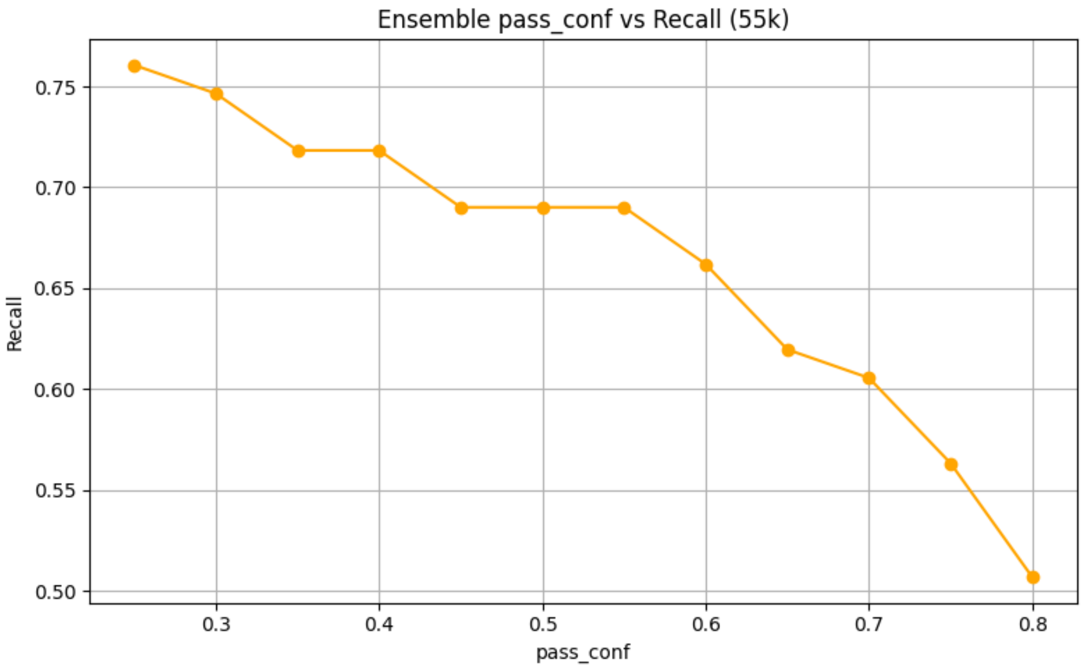


Figure 3.7: Recall as a function of ensemble confidence threshold.

large-scale exposure analysis.

3.6 Summary

This section concludes the model evaluation and selection process prior to large-scale real-world application. Through a sequence of controlled experiments, the proposed framework progressively transitions from instance-level evaluation to frame-level analysis and finally to large-scale robustness cross-check on raw video data. This staged evaluation design is intended to bridge the gap between benchmark-oriented model comparison and practical deployment under real-world conditions (Chung et al., 2019; Hawke, 2025).

The results demonstrate that while multiple models perform competitively under balanced validation settings, their behaviour diverges under highly imbalanced, real-world conditions. By integrating ensemble learning and confidence threshold calibration, a stable operating configuration is identified that maintains competitive detection performance while effectively suppressing false positives

Importantly, the large-scale raw-frame cross-check confirms that the selected ensemble generalises beyond controlled validation data and remains well-behaved under realistic deployment conditions, where brand exposures are sparse and false positives can otherwise dominate the analysis. This confirms the suitability of the selected configuration for downstream exposure analysis.

With the model configuration finalised, the next stage of the study applies the selected pipeline to real-world screen recordings to quantify brand exposure patterns in natural viewing environments. The following chapter therefore focuses on the end-to-end application of the proposed detection pipeline and the analysis of exposure results.

Chapter 4

Pipeline for Kids Online Data

Following the model selection and evaluation procedures described in Chapter 3, this chapter focuses on the practical application of the proposed detection pipeline. While the previous chapter established the effectiveness of individual models and ensemble configurations under controlled, ground-truth-based conditions, the objective here is to demonstrate how the finalised pipeline can be deployed on real-world, unlabelled screen-recorded data.

This chapter is organised into two main parts. The first part describes the construction and integration of the full detection pipeline and presents a functional validation using annotated data. This stage serves to verify that the pipeline behaves as intended and that the integration of visual, textual, and semantic cues operates in a stable and coherent manner.

The second part applies the validated pipeline to the Kids Online screen-recorded data, where no ground truth annotations are available. Rather than evaluating detection performance, this stage focuses on practical deployment, illustrating how the pipeline can be used to identify and summarise potential brand exposure in children's viewed media under real-world conditions.

Through this two-stage design, the chapter bridges controlled experimental validation and real-world application, illustrating how the proposed system can be operationalised for large-scale exposure analysis. This application-oriented framing emphasises feasibility, robustness, and interpretability over benchmark accuracy, aligning with the broader objectives of the Kids Online project.

4.1 Pipeline

The proposed pipeline operates at the frame level and integrates three complementary components: visual logo detection using an ensemble of object detection models, text-based detection based on optical character recognition (OCR), and semantic scene understanding using a vision–language model. Each component captures different aspects of brand-related information and produces an independent signal for exposure detection. These signals are combined to generate a final exposure decision for each target brand on a per-frame basis.

Visual detection serves as the primary source of evidence for brand exposure. Textual and semantic signals are incorporated in a supporting role through a rule-based confirmation and bonus integration mechanism. Specifically, OCR and CLIP outputs do not independently trigger exposure decisions. Instead, they are used to reinforce visual detections when corresponding evidence is present. The detailed frame-level decision logic governing this multimodal confirmation and bonus integration mechanism is provided in Appendix A.1. This design reflects the central role of explicit visual logo localisation as the core evidence source in object detection pipelines and applied exposure analysis (Benjelloun et al., 2020; Hawke, 2025).

For OCR, textual content is extracted from the full frame and matched against predefined brand keywords using fuzzy matching. When a high-confidence text match is detected and a corresponding visual detection exists, a direct reinforcement is applied. For lower-confidence matches, a small bonus is added to the visual score to strengthen borderline detections. The use of OCR as a complementary signal has been shown to be effective in disambiguating visually similar objects and reducing false positives in detection pipelines involving branded content (Yumang et al., 2022; Zong et al., 2025).

Similarly, CLIP is used to assess semantic confidence between the frame and predefined brand-related prompts. When a high-confidence semantic match is observed in the presence of a corresponding visual detection, a direct reinforcement is applied to confirm the candidate exposure. For lower-confidence semantic matches, a limited bonus is added to the visual score to support borderline cases. As with OCR, semantic cues from CLIP are not used as a standalone decision signal, but serve as a complementary confirmation mechanism that strengthens detection confidence under partial or ambiguous visual evidence. This design choice is motivated by prior findings that vision–language models have been shown to provide effective contextual validation in visually noisy or unconstrained environments (Radford et al., 2021; Cheng et al., 2024).

The contributions from OCR and CLIP are intentionally constrained to ensure that visual evidence remains the dominant factor in the decision process. A final exposure decision is made only when the aggregated score exceeds the predefined threshold determined in Chapter 3. This conservative integration strategy enables the pipeline to leverage multi-modal information while maintaining robustness against noise and false positives in real-world screen-recorded data, a key requirement for exposure-oriented analysis under extreme class imbalance.

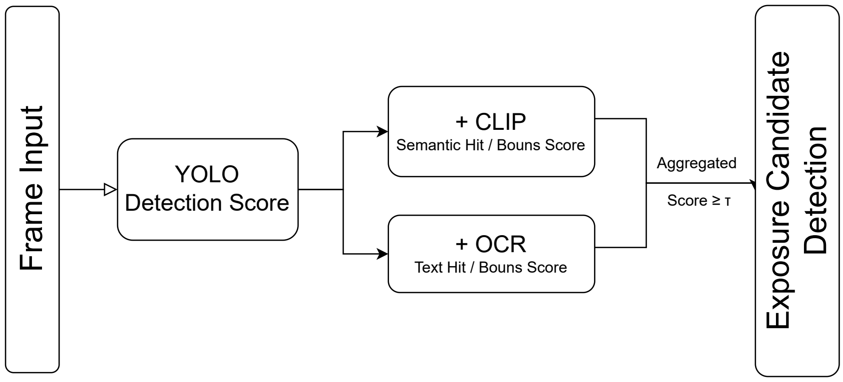


Figure 4.1: Overview of the proposed multi-stage brand exposure detection pipeline

4.1.1 Visual Detection and Ensemble Scoring

Visual brand detection in the proposed pipeline is performed using an ensemble of object detection models, specifically YOLOv11 and YOLO-World. These models are selected based on the evaluation results reported in Chapter 3 and are used jointly to capture complementary detection characteristics under diverse visual conditions, including closed-set logo detection accuracy and open-vocabulary robustness.

For each input frame, both models are applied independently to generate brand-related detections. Consistent with the evaluation setting described in Chapter 3, detections are first consolidated at the model level. When multiple bounding boxes corresponding to the same brand are produced by a single model within a frame, only the highest confidence score is retained to represent that brand. This ensures that each model contributes at most one score per brand per frame and prevents repeated detections of the same object from inflating exposure estimates.

After this per-model consolidation, the outputs of the individual models are combined using a weighted ensemble strategy. Specifically, the retained confidence score from each model is multiplied by a predefined model weight, and the weighted scores are summed to produce a single visual confidence value for each brand in the frame. The model weights are fixed and determined based on the evaluation and threshold selection procedures described in Chapter 3. This formulation allows the pipeline to leverage the complementary strengths of the two detectors while maintaining a stable and interpretable scoring mechanism. In the final deployment configuration, the ensemble assigns a higher weight to YOLOv11 than to YOLO-World, following a weighting of 0.6 for YOLOv11 and 0.4 for YOLO-World respectively.

The resulting visual score represents the final visual evidence of brand presence for each brand at the frame level. This score is subsequently used in the downstream exposure analysis together with textual and semantic cues. By structuring the ensemble at the score level rather than the bounding-box level, the pipeline prioritises stability and interpretability over fine-grained localisation accuracy, which is consistent with the requirements of exposure-oriented analysis under highly imbalanced conditions.

4.1.2 Text-based Detection (OCR)

Text-based detection is incorporated into the pipeline as a supplementary signal to support visual logo detection. Optical character recognition (OCR) is implemented using EasyOCR and is applied to the full frame rather than to cropped regions. This design choice ensures that textual cues appearing anywhere in the screen, such as brand names, packaging text, or on-screen advertisements, can be captured, which is particularly important in screen-recorded data where text placement is highly variable (Yumang et al., 2022).

In this study, OCR is not treated as an independent detection mechanism. Instead, it functions strictly as a supporting component that is evaluated only when a corresponding visual detection is present. Text-only detections are therefore unable to generate exposure events on their own.

Extracted text is matched against predefined brand keywords using fuzzy string matching, and two confidence levels are defined. When the fuzzy match score exceeds a high threshold (0.90) and a corresponding visual detection exists, textual evidence is considered sufficiently strong to confirm the visual detection. A strong textual confirmation bonus (+1.0) is applied, which guarantees a positive exposure decision under the fixed multimodal threshold while remaining within the unified fusion framework.

For lower-confidence textual matches that exceed a secondary threshold (0.70), OCR does not guarantee exposure. Instead, a weak OCR bonus is applied, contributing a small additive score to the corresponding visual detection. This mechanism allows textual cues to reinforce borderline visual evidence while preserving conservative detection behaviour. Accordingly, the contribution of OCR is intentionally constrained to ensure that visual evidence remains the dominant factor in the decision process.

4.1.3 Semantic Detection (CLIP)

Semantic information is derived from a vision–language model (CLIP) to support brand exposure detection. CLIP is used to evaluate the semantic confidence between an input frame and a set of predefined textual prompts representing brand-related, fast food-related, and non-relevant concepts. In this study, CLIP is not employed as an independent detection mechanism and is evaluated only when a corresponding visual detection is present. CLIP-based signals are therefore unable to generate exposure events in the absence of visual evidence, ensuring that semantic reasoning is always grounded in explicit visual localisation.

A structured prompt set is defined to guide semantic matching, including explicit brand prompts, generic food-related prompts, and contextual negative prompts representing non-food or irrelevant activities. This contrastive prompt design enables the model to distinguish between food-related scenes and common non-relevant screen content such as application interfaces, messaging, or general browsing, and has been shown to improve semantic discrimination in open-vocabulary and vision–language settings (Radford et al., 2021; Chhipa et al., 2024).

For each frame, CLIP prompt-match confidence scores are computed and the most confident semantic match is identified. When a strong brand-related semantic match is detected (prompt-match confidence ≥ 0.90) and a corresponding visual detection exists, semantic evidence is considered sufficiently strong to confirm the visual detection. A strong semantic confirmation bonus (+1.0) is applied, which guarantees a positive exposure decision under the fixed multimodal threshold while remaining within the unified fusion framework.

For lower-confidence semantic matches that exceed a secondary threshold (0.50), CLIP does not guarantee exposure. Instead, a weak semantic bonus is applied, contributing a small additive score to the corresponding visual detection. Generic food-related prompts contribute only weak reinforcement and are applied conservatively to support borderline visual evidence without inducing over-generalisation.

The contribution of CLIP is intentionally constrained to ensure that semantic cues cannot override visual evidence in the absence of explicit localisation. Details of the CLIP prompt categories, prompt-to-brand mapping rules, and semantic reinforcement logic are documented in Appendix A.2.

4.1.4 Score Aggregation and Exposure Decision

The final exposure decision is made by aggregating the outputs from the visual, textual, and semantic components according to the pipeline rules described above. Visual detection serves as the primary evidence source, while OCR and CLIP contribute supporting signals that are evaluated only in conjunction with visual detections.

For each frame and each brand, a final confidence score is computed based on the visual detection score and any applicable bonus contributions from OCR and CLIP. In the general case, an exposure is recorded only when the aggregated score exceeds the predefined decision threshold established in Chapter 3.

In addition to score-based aggregation, strong textual or semantic evidence provides decisive reinforcement when accompanied by a corresponding visual detection. Specifically, high-confidence OCR or CLIP matches are treated as strong reinforcement signals that substantially increase the multimodal confidence score. Given an existing visual detection, this strong bonus deterministically elevates the final score above the fixed threshold, guaranteeing a positive exposure decision while remaining within the unified fusion framework. This design preserves conservative behaviour in ambiguous cases while allowing highly reliable multimodal evidence to dominate the final decision.

Overall, this formulation ensures that exposure decisions are driven primarily by explicit visual evidence, while allowing textual and semantic cues to reinforce or decisively support detections when appropriate. The resulting decision process is conservative, interpretable, and well suited for large-scale analysis of unlabelled screen-recorded data under extreme class imbalance.

4.2 Pipeline Robustness Cross-check with Ground Truth

This section examines the behaviour of the final detection pipeline under realistic operating conditions. Using the final configuration selected in Chapter 3, the complete pipeline, including ensemble filtering, OCR, and CLIP-based semantic cues, is applied to two previously introduced datasets: (1) the frame-level validation set, originally used for ensemble selection, and (2) the large-scale raw-frame dataset used for robustness analysis.

Across both datasets, the proposed pipeline exhibits a clear reduction in false positive rates while maintaining, and in some cases slightly improving, stable detection behaviour relative to the single-model detectors (YOLOv11 and YOLO-World). This improvement is primarily attributable to the ensemble mechanism and confidence-based filtering, which suppress detections that are not consistently supported across models. Similar stabilising effects of ensemble strategies under noisy and heterogeneous conditions have been reported in recent object detection studies (Jiang and Zhong, 2025). This property is particularly important in large-scale, highly imbalanced exposure analysis, where even small false positive rates can substantially distort downstream estimates (Russell, 2002).

On the frame-level validation set, the inclusion of OCR and CLIP leads to a small increase in recall, while the false positive rate remains effectively unchanged. This indicates that textual and semantic components primarily function as auxiliary recall-enhancement signals rather than as independent filtering mechanisms (Table 4.1).

Table 4.1: Pipeline-level robustness cross-check on the frame-level validation dataset

Model / Configuration	Conf.	Precision	Recall	F1-score	FP rate
YOLOv11 Best F1	0.55	0.9631	0.7986	0.8732	0.016
YOLO-World Best F1	0.30	0.9540	0.8579	0.9034	0.022
Ensemble (v11 + WD)	0.50	0.9954	0.7806	0.8750	0.002
Full pipeline (Ensemble + OCR + CLIP)	0.50	0.9955	0.7914	0.8818	0.002

A more pronounced effect is observed on the large-scale raw-frame dataset. In this setting, the ensemble configuration substantially suppresses false positives, reducing the false positive rate to a stable low level. Building upon this baseline, the inclusion of OCR and CLIP does not introduce additional false positives, but instead recovers a small number of borderline true positives, leading to a modest but consistent improvement in recall (Table 4.2).

This behaviour highlights a clear division of roles within the pipeline. The ensemble mechanism primarily functions as a false-positive suppression layer, removing low-confidence or inconsistent detections that disproportionately inflate false positive counts. In contrast, OCR and CLIP act as selective recall-enhancement components, activating only in ambiguous cases where supporting textual or semantic cues are

Table 4.2: Pipeline-level robustness cross-check on the large-scale raw-frame dataset

Model / Configuration	Conf.	Precision	Recall	F1-score	FP rate
YOLOv11 Best F1	0.55	0.0600	0.7200	0.1100	0.015
YOLO-World Best F1	0.30	0.0400	0.7300	0.0800	0.021
Ensemble (v11 + WD)	0.50	0.1800	0.7000	0.2900	0.004
Full pipeline (Ensemble + OCR + CLIP)	0.50	0.1835	0.7083	0.2914	0.004

present.

As a result, recall is slightly increased without compromising false positive control, while precision remains stable with a marginal improvement. This complementary behaviour indicates that the proposed pipeline achieves improved robustness and practical reliability over ensemble-only configurations when applied to large-scale, highly imbalanced real-world data subject to distribution shift.

4.3 Application to Kids Online Data

Following the cross-check of the proposed pipeline on annotated datasets, the method is further applied to real-world screen-recorded data collected as part of the Kids Online project. In contrast to the earlier evaluation stages, this dataset does not contain ground truth annotations, and the presence of brand exposure is unknown prior to analysis.

The raw data consists of long-duration screen recordings capturing a wide range of everyday digital activities, including application navigation, scrolling behaviour, messaging, and video playback. Although the primary interest of this study lies in brand exposure occurring during video viewing, the recorded frames are not limited to video content. Instead, they frequently include interface elements, system transitions, and other non-video interactions.

Such non-video frames can give rise to false positive detections, particularly when interface layouts or background graphics exhibit visual patterns resembling brand-related elements. However, these detections do not correspond to meaningful exposure events in the context of media consumption. Rather than manually excluding these frames, the pipeline is intentionally applied to the complete, unfiltered recordings. This design choice reflects realistic deployment conditions and avoids introducing subjective bias through manual frame selection.

After candidate exposure frames are identified, a limited manual verification step is applied solely for the purpose of result interpretation. This step consists of two components: (1) the exclusion of non-video frames that do not correspond to meaningful media consumption events, and (2) limited manual verification to remove residual false positives.

This manual verification reflects the semi-automated nature of the proposed pipeline. Although the selected ensemble configuration achieves a low false positive rate (approximately 0.3% to 0.5%), false detections may still occur when applied to long-duration, large-scale recordings. Importantly, this verification is performed only after all automated detection stages have completed and does not modify model predictions, confidence thresholds, or pipeline behaviour. Its sole purpose is to support accurate interpretation of exposure results while preserving the integrity and reproducibility of the automated pipeline.

Due to time and resource constraints, the analysis in this study focuses on data collected in 2023 and 2024 from a single randomly selected participating school. The aim of this stage is not to derive definitive exposure statistics, but to evaluate whether the proposed pipeline can operate reliably on real-world data, produce interpretable outputs, and support scalable deployment in future large-scale studies.

4.3.1 Preprocessing Strategy

To enable efficient processing of long screen-recorded videos while preserving relevant visual information, a multi-stage preprocessing strategy is adopted. The primary objectives of this stage are to reduce temporal redundancy, minimise unnecessary computation, and retain sufficient visual coverage for downstream exposure analysis.

Rather than applying uniform frame sampling, the preprocessing pipeline adopts a content-aware extraction strategy specifically designed for screen-recorded data. This approach accounts for the highly irregular temporal structure of screen recordings, which typically contain long periods of static content interspersed with short bursts of visual change. Fixed-rate sampling in such settings often leads to excessive redundancy or missed visual events (Chung et al., 2019).

The adopted preprocessing strategy is adapted from the frame extraction framework proposed in prior work by Hawke (2025), which was developed for analysing long-form screen recordings in the context of the Kids Online project. The method has been shown to achieve an effective balance between temporal representativeness and computational efficiency by selectively retaining visually informative frames while discarding redundant ones.

By applying this content-aware preprocessing scheme, the pipeline ensures that downstream exposure analysis is performed on a compact yet representative set of frames. This design significantly reduces processing cost while preserving the visual cues necessary for reliable brand exposure detection in real-world screen-recorded data, where exposure events are sparse and irregularly distributed.

4.3.2 Frame Extraction Rules

The extraction procedure follows three main rules:

(1) Redundant Frame Filtering Frames are discarded when minimal visual change is observed relative to the most recently retained frame within a short temporal window. Specifically, if the time elapsed since the last retained frame is less than two seconds, a frame is excluded when the absolute pixel-wise difference falls below a fixed threshold of 5000, indicating near-identical visual content. This rule removes redundant frames caused by static screens, prolonged application states, or slow interface interactions, which are common in screen-recorded data.

(2) Temporal Coverage with Change-triggered Retention To ensure sufficient temporal coverage without excessive sampling, at least one frame is retained every two seconds throughout the video. In addition, frames exhibiting substantial visual change may be retained within this interval, subject to a minimum separation of one second between successive change-triggered frames. Together, these rules impose a conditional upper bound of approximately two frames per second. The first and final frames of each video are always preserved to ensure complete temporal coverage. Change-triggered retention operates independently of the periodic sampling rule and does not reset the two-second timer.

This design captures short-lived visual events while preventing an excessive number of frames from being generated in visually dynamic scenes, a common challenge in long-duration video analysis (Chung et al., 2019).

(3) Image Normalisation All extracted frames are downsampled to a maximum long edge of 800 pixels (no upsampling is applied) while preserving the original aspect ratio. This normalisation step reduces computational cost and ensures consistent input resolution for downstream detection models, supporting stable inference behaviour across heterogeneous devices and recording resolutions (Benjelloun et al., 2020).

Overall, this rule-based extraction strategy, originally introduced by Hawke (2025), has been shown to be effective for processing long-form screen-recorded data, where visual content is sparse, repetitive, and irregularly distributed over time. The same strategy is adopted in this study to ensure methodological consistency and computational efficiency across the analysis pipeline. The corresponding pseudo-code implementation of the frame extraction procedure is provided in Appendix A.3.

4.3.3 Application to Kids Online Data without Ground Truth

This section focuses on the application of the proposed pipeline to real-world screen-recorded data. Unlike the earlier experimental stages, no performance evaluation or accuracy assessment is conducted at this stage by design, as ground truth annotations are not available for the Kids Online data. In the absence of verified labels, the analysis is necessarily descriptive rather than evaluative.

The analysis is performed on a large-scale real-world dataset comprising approximately 191 hours, 36 minutes, and 21 seconds of screen-recorded video, corresponding to a total of 734,432 extracted frames. The pipeline is applied directly to the complete set of extracted frames. All detected instances are therefore treated as candidate exposure events. The resulting outputs consist of frame-level exposure records, including detected brand categories, associated confidence scores, and temporal information such as timestamps. These structured outputs provide the basis for subsequent descriptive analysis of exposure patterns.

Due to the absence of ground truth annotations, no attempt is made to compute performance metrics such as precision, recall, or accuracy. Instead, the focus is placed on the practical usability of the pipeline, the interpretability of its outputs, and its ability to process large volumes of real-world data in a stable and scalable manner.

At the final stage of analysis, a limited manual verification step is applied to the output results to remove a small number of residual false positives and clearly non-target frames. This manual verification step is performed solely for result interpretation and presentation purposes and is not used to retroactively validate or correct model performance. This form of limited human-in-the-loop filtering is not unique to the present study; similar strategies have been used in applied exposure studies to balance automation with interpretability and quality control (Smith et al., 2019; Hawke, 2025). The results therefore indicate that the proposed pipeline significantly reduces human effort while maintaining acceptable reliability for large-scale exploratory exposure analysis.

Overall, this application-oriented analysis illustrates that the proposed system can operate effectively under real-world conditions, producing structured and interpretable exposure outputs with minimal human intervention. Rather than serving as a benchmarked detection model, the pipeline is intended as a practical and scalable tool for exploratory studies of brand exposure in naturalistic digital media environments.

4.3.4 Results

This section presents the results obtained from applying the proposed pipeline to real-world screen-recorded data. The analysis focuses on two aspects: (1) the characteristics of detected brand exposure events, and (2) the overall operational performance of the pipeline when applied at scale.

Exposure Definition

In this study, a deliberately simplified exposure definition is adopted. An exposure event is defined based on temporal continuity: if detections of the same brand occur within a time gap of no more than two seconds, they are grouped as a single exposure event, regardless of whether the detections occur within continuous playback or across brief temporal gaps between short video clips within the same recording. No spatial consistency or bounding box overlap is required for an exposure to be counted.

This simplified definition is used because detailed exposure modelling is not within the scope of this research. The primary objectives of this study are: (1) to examine the role of semantic representations, such as those derived from YOLO-World and CLIP, in supporting pixel-level logo and brand detection within screen-recorded media; and (2) to develop and evaluate a semi-automated multi-model pipeline that integrates YOLO-based detection, OCR, and CLIP for large-scale brand exposure analysis in children's viewed media.

Accordingly, the exposure rule is applied only as a lightweight aggregation mechanism to summarise frame-level detections into interpretable exposure units for reporting and downstream use. The intention is not to derive precise behavioural exposure metrics, but to provide structured outputs that can be utilised by future studies or domain experts who wish to conduct more detailed exposure analysis.

Furthermore, only detections occurring during video-watching activities are included in the exposure analysis. Frames associated with application home pages or food ordering interfaces are excluded, even when brand logos are correctly detected. This ensures that the reported exposure events reflect media consumption contexts rather than incidental interface interactions.

Exposure Summary

Across the analysed dataset, brand exposure events were detected for multiple fast-food brands, including McDonald's, KFC, Burger King, Domino's Pizza, and Pizza Hut. Exposure events were observed across different devices (phone, laptop, and Desktop/tablet), time periods (morning, afternoon, night), and recording sessions.

The results indicate that brand exposure events are generally short in duration and sparse in occurrence. Across all brands and devices, most exposure events last only a few seconds, with the majority of exposures falling below two seconds on average.

Even for brands with relatively higher exposure frequency, such as McDonald's and Domino's Pizza, individual exposure events remain brief and fragmented, rather than forming prolonged continuous appearances. This pattern is consistent across device types and time periods, suggesting that brand appearances in naturalistic screen use are typically embedded within broader viewing activities rather than constituting sustained visual focus (Table 4.3).

Table 4.3: Summary of detected brand exposure events by brand across the 2023–2024 Kids Online dataset.

Brand	No. exposures	Sum of duration (s)	Average duration (s)
Burger King	17	57	3.35
Domino's Pizza	24	41	1.71
KFC	11	15	1.36
McDonald's	92	156	1.70
Pizza Hut	3	2	0.67
Grand Total	147	271	1.84

Figure 4.2 compares the total brand exposure length (in seconds) across major fast-food brands between 2023 and 2024. Overall, McDonald's shows the longest cumulative exposure in both years, with a substantial increase in 2024, indicating more frequent or longer total appearances across the analysed recordings. Burger King and Domino's Pizza display moderate exposure lengths, with year-to-year variation but no dramatic change. KFC and Pizza Hut exhibit relatively low total exposure, remaining marginal compared with other brands. From a temporal perspective, the chart highlights that differences between brands are more pronounced than differences between years. This suggests that brand visibility is unevenly distributed, with a small number of dominant brands accounting for most cumulative exposure time, while others appear only

sporadically.

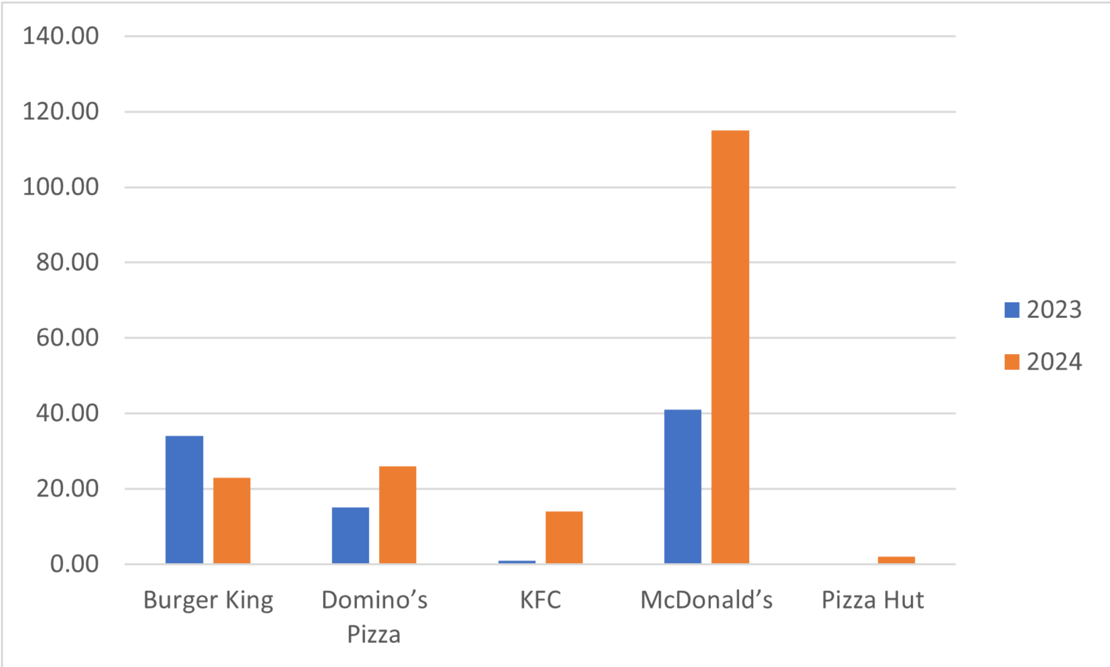


Figure 4.2: Total detected brand exposure duration (in seconds) by year (2023–2024).

Figure 4.3 summarises the distribution of detected brand exposure events across years. As shown in the stacked bar chart, McDonald's accounts for the largest proportion of exposure events in both 2023 and 2024, substantially exceeding other brands. KFC, Domino's Pizza and Burger King appear less frequently, while Pizza Hut contributes only a small number of exposure instances. Although the total amount of processed screen-recorded data is comparable across the two years (97:01:30 in 2023 and 94:34:51 in 2024), the number of detected exposure events is notably higher in 2024. This indicates an overall increase in exposure frequency rather than an artefact of longer recording duration, suggesting a higher density of brand appearances within a similar amount of viewing time in the later data.

Figure 4.4(a) shows that exposures occur most frequently during morning and night. Figure 4.4(b) shows that mobile phones dominate exposure occurrences, accounting for the majority of detected events. Laptop-based exposure is less frequent, while desktop or tablet usage contributes only a small proportion. This reflects the central role of mobile devices in everyday media consumption among children. Finally, Figure 4.4(c) shows a highly uneven distribution of exposure across brands. McDonald's alone represents the majority of detected exposures, whereas other brands appear sporadically.

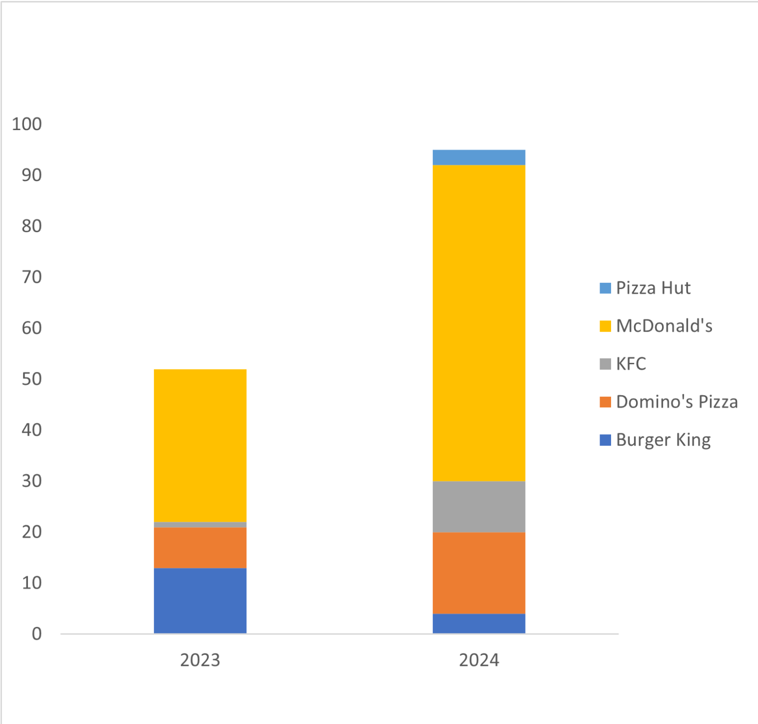


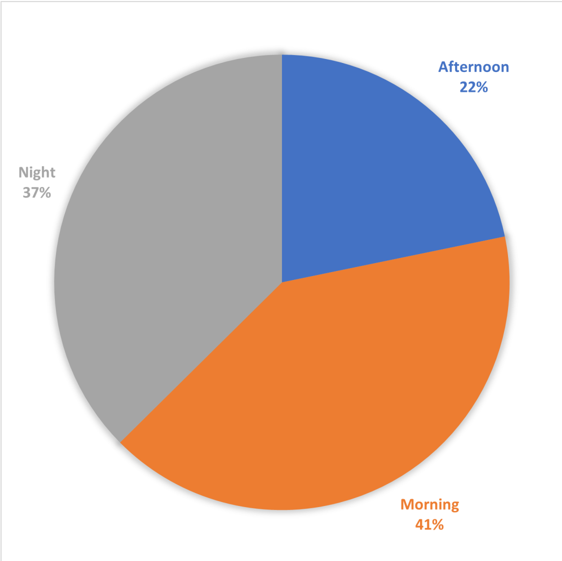
Figure 4.3: Frequency of detected brand exposure events by year (2023–2024).

This skewed distribution reinforces the observation that real-world brand exposure is highly concentrated and episodic rather than evenly distributed.

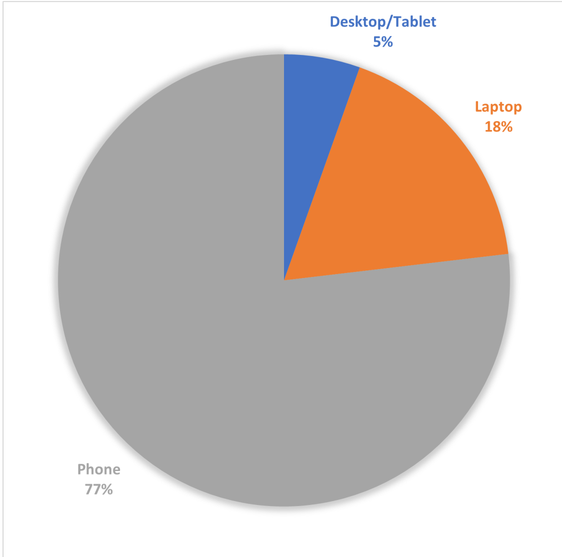
Pipeline Performance Summary

In total, the pipeline processed 734,423 frames, corresponding to 191 hours, 36 minutes, and 21 seconds of screen-recorded video. Across this dataset, 483 final exposure frames were identified following automated detection and limited manual verification, alongside 2,440 false positive detections, resulting in an overall false positive rate of approximately 0.33%.

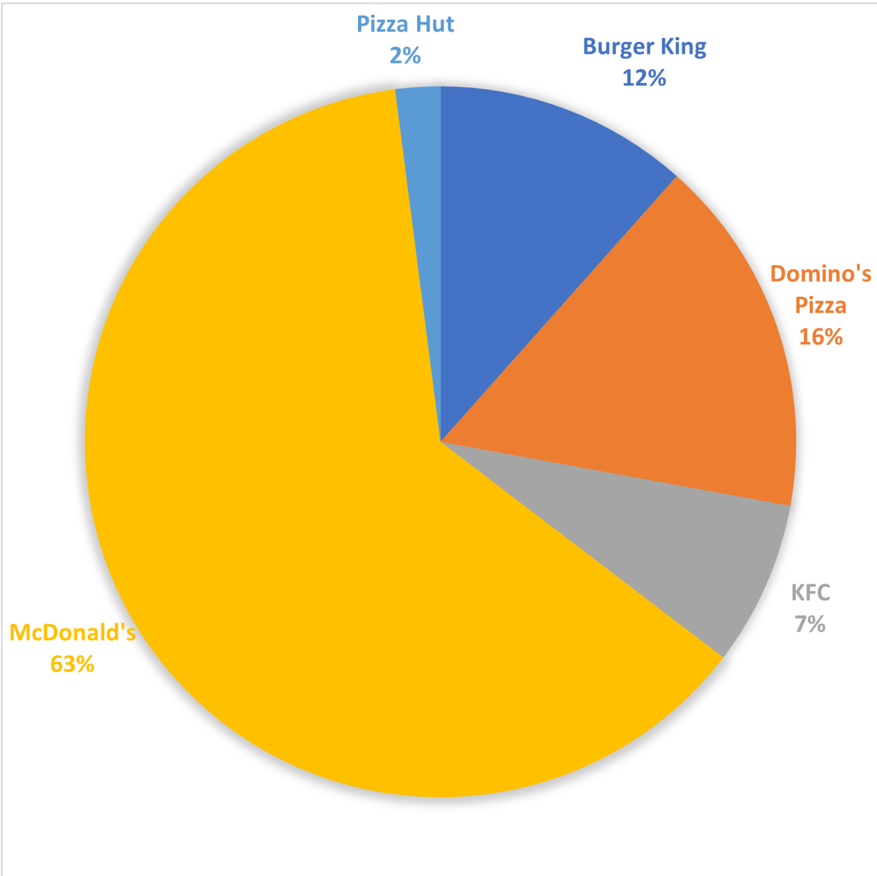
Although the false positive rate is low in relative terms, the absolute number of false detections is non-trivial due to the scale of the data. As noted above, the limited manual verification step was applied at the final stage to remove such cases for result interpretation. Importantly, even with this verification step, the workload remains substantially lower than manual inspection of the full dataset. Reviewing over 190 hours of video or more than 730,000 frames manually would be infeasible in practice. The proposed pipeline therefore achieves a substantial reduction in human effort while enabling practical identification of candidate exposure events.



(a) Time of day



(b) Device type



(c) Brand composition

Figure 4.4: Distribution of detected brand exposure events by (a) time of day, (b) device type, and (c) brand composition across all analysed recordings.

To further illustrate the operational behaviour of the pipeline, Figures 4.5 and Figures 4.6 present representative qualitative examples observed during large-scale deployment. These examples highlight typical cases where auxiliary components recover true positives that would otherwise be missed by visual detection alone, as well as residual failure cases that motivate the need for limited manual verification.

Overall, the results demonstrate that the proposed system can operate stably on large-scale, real-world screen recordings, producing structured exposure outputs that support exploratory analysis of brand exposure patterns in naturalistic digital media environments.

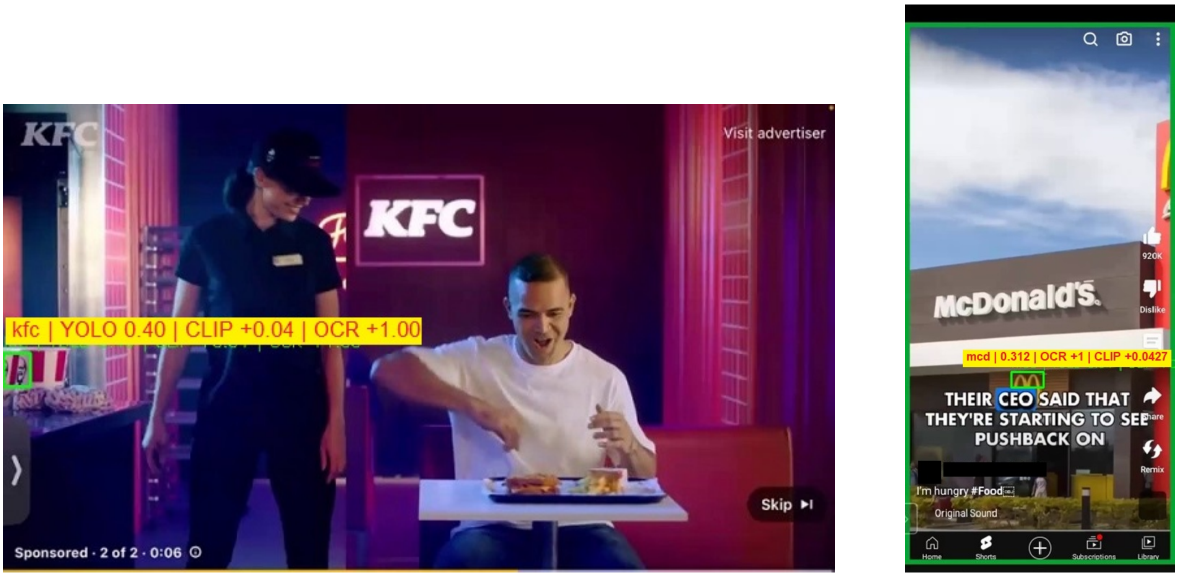
4.3.5 Discussion: Real-world Application Behaviour & Limitations

The application of the proposed pipeline to real-world screen-recorded data reveals several consistent behavioural patterns. Across both years, brand exposure events are generally short in duration and sparse in occurrence, with most exposures lasting only a few seconds. Exposure events are predominantly observed during mobile phone usage and occur more frequently during morning and night-time viewing periods, as observed in the analysed recordings.

Despite the total amount of processed data being comparable between the two years, a noticeably higher number of exposure events is observed in 2024. This indicates that the increase in exposure frequency is not an artefact of longer recording duration, but rather reflects a higher density of brand appearances within a similar amount of viewing time.

It is important to note that this analysis is based on data collected from a single participating school over a two-year period. Although the total number of processed frames is substantial, the dataset represents a limited sample in terms of user population, viewing habits, and contextual diversity. As a result, the observed exposure patterns should be interpreted as illustrative rather than representative of broader populations.

Furthermore, exposure events in this study are defined using a simplified temporal rule without modelling visual salience, object persistence, or user attention. The purpose of this design choice is not to estimate perceptual impact, but to evaluate whether a unified detection pipeline can operate reliably on large-scale, uncurated screen-recorded data. The results therefore demonstrate the practical feasibility and



(a) True positive cases recovered by OCR where visual-only detection failed, observed consistently across desktop and mobile recordings.

(b) True positive cases recovered through CLIP-based semantic reinforcement under weak visual evidence on both desktop and mobile platforms.

Figure 4.5: Representative qualitative examples illustrating recovery behaviour of the proposed pipeline during real-world deployment. All examples are anonymised.

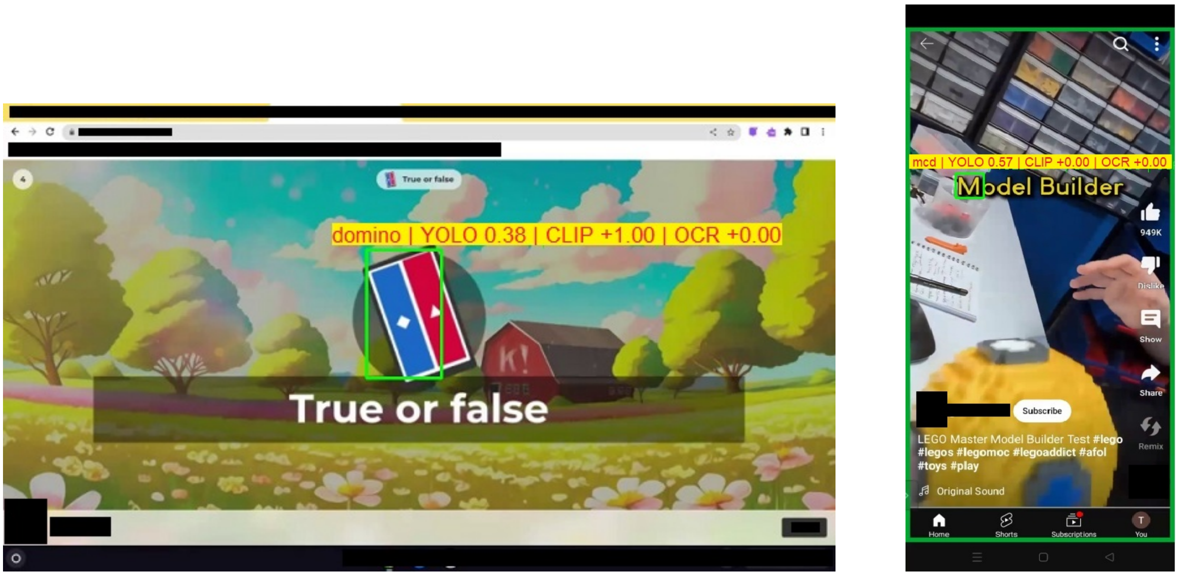


Figure 4.6: Representative qualitative examples illustrating residual failure modes observed during pipeline deployment, motivating the need for limited manual verification under large-scale application. All examples are anonymised.

behavioural characteristics of the proposed system, rather than providing definitive measurements of advertising exposure or user-level impact.

Chapter 5

Discussion and Conclusion

5.1 Discussion

5.1.1 Limitations

Despite the overall effectiveness of the proposed pipeline, several limitations should be acknowledged. These limitations primarily reflect deliberate design trade-offs made to prioritise system-level feasibility and scalability.

- **Limited optimisation of auxiliary modules**: The OCR and CLIP components were incorporated as supporting modules and were not extensively tuned. The primary focus of this study was the development of a stable ensemble framework based on YOLOv11 and YOLO-World, rather than exhaustive optimisation of individual pipeline stages.

- **Constrained effectiveness of CLIP under zero-shot settings**: Although CLIP provides semantic reasoning capability in principle (Radford et al., 2021), its contribution under the current zero-shot and full-frame configuration was limited. In some cases, CLIP detections overlapped with YOLO-World outputs or introduced additional false positives, indicating that more systematic prompt design or region-level reasoning would be required (Chhipa et al., 2024; Zong et al., 2025).

- **Simplified false positive handling**: False positives were mainly controlled through ensemble filtering, confidence thresholds, and manually selected negative samples. More advanced false positive suppression strategies, such as confidence reweighting methods, were not implemented (Wu et al., 2026).

- **Lack of temporal modelling**: The pipeline operates at the frame level and does not explicitly model temporal continuity. As a result, exposure events are treated as independent observations rather than temporally coherent segments (Chung et al., 2019).

These limitations do not undermine the feasibility of the proposed pipeline but define clear boundaries for its current scope.

5.1.2 Future Work

Several directions may be explored to further strengthen the proposed pipeline:

- **Optimisation of OCR and CLIP components**: Future work may incorporate image-level preprocessing for OCR, such as resolution adjustment or sharpness enhancement, which has been shown to improve text recognition accuracy in video-derived imagery (Yin et al., 2016; Li et al., 2022). More systematic prompt engineering or region-level reasoning may also enable CLIP to better exploit its semantic reasoning capabilities (Radford et al., 2021; Chhipa et al., 2024).

- **Advanced false positive suppression strategies**: More principled approaches to false positive control, such as confidence reweighting or improved handling of negative samples, could be integrated to further improve robustness under highly imbalanced conditions (Wu et al., 2026).

- **Incorporation of temporal modelling**: Temporal smoothing or sequence-based aggregation could be introduced to better capture exposure duration and persistence, improving robustness and interpretability in video-based detection tasks (Chung et al., 2019).

- **Exploration of alternative detection architectures and efficiency evaluation**: Future studies may compare the current YOLO-based framework with grounding-based models, such as Grounding DINO combined with segmentation approaches, to evaluate trade-offs between detection accuracy and deployability (Ren et al., 2024b; Sun et al., 2025). Explicit efficiency evaluation and model compression techniques may also be explored to better characterise large-scale deployment feasibility (Liqun and Lei, 2023).

5.2 Conclusion

This study developed and evaluated a unified pipeline for brand exposure detection in screen-recorded data, with the primary objective of assessing its feasibility as a practical analytical tool. By integrating YOLO-based detection, OCR, and semantic reasoning components, the proposed pipeline demonstrated the ability to process large volumes of unstructured screen recordings and to identify potential brand exposure events under real-world conditions. This system-oriented contribution aligns with the needs of applied exposure research, where scalability and robustness are often prioritised over isolated model performance.

Rather than pursuing maximal detection accuracy under controlled benchmarks, this study focused on robustness, scalability, and applicability to real-world data. The results show that the pipeline can effectively support large-scale exposure screening, even in the absence of frame-level ground truth annotations.

The observed sparsity of exposure events and the behaviour of the system across different data subsets further illustrate the challenges inherent in analysing real-world screen-recorded content. Such sparsity is consistent with prior findings that commercially relevant content typically constitutes only a small fraction of total viewing time in children's digital media environments (Smith et al., 2019). These observations highlight the importance of false positive control and conservative decision strategies when applying automated detection systems at scale.

At the same time, the study highlights several limitations related to model optimisation, false positive handling, and semantic reasoning, which point to clear directions for future improvement. These limitations do not undermine the validity of the proposed approach, but instead reflect the trade-offs involved in designing a practical and extensible system for real-world deployment, particularly under conditions of extreme class imbalance and visual variability (Chhipa et al., 2024).

Overall, this work contributes a reproducible and scalable framework for brand exposure analysis and provides a foundation for future research aimed at improving automated exposure measurement in complex, real-world digital environments. By demonstrating how object detection, textual cues, and semantic reasoning can be integrated in a conservative and interpretable manner, the study supports the broader goal of advancing data-driven analysis of children's media exposure at scale.

Appendix A

Supplementary Material

A.1 Multimodal Confirmation Logic

The following pseudo-code summarises the frame-level decision logic used for integrating OCR and CLIP signals into the proposed detection pipeline. The logic is applied only when a corresponding visual detection is present.

```
Algorithm: Multimodal Confirmation Logic (OCR / CLIP)

For each frame:
    If visual detection exists for brand b:
        If OCR confidence(b) >= strong_threshold:
            Add strong OCR bonus to visual score
        Else if OCR confidence(b) >= weak_threshold:
            Add small OCR bonus to visual score

        If CLIP confidence(b) >= strong_threshold:
            Add strong CLIP bonus to visual score
        Else if CLIP confidence(b) >= weak_threshold:
            Add small CLIP bonus to visual score

    Exposure is recorded if:
        - the final aggregated score >= decision threshold
```

A.2 CLIP Prompt Design and Semantic Reinforcement Rules

This section documents the prompt design and semantic reinforcement strategy adopted for the CLIP-based enhancement module. The prompts and decision rules are disclosed to support transparency and reproducibility. CLIP is applied only as a supplementary semantic signal and does not operate as an independent detector.

A.2.1 CLIP Prompt Categories

The prompt set consists of three categories: direct brand prompts, generic food-related prompts, and negative prompts representing common non-brand screen activities.

Direct brand prompts

- McDonalds
- Burger King
- KFC
- Pizza Hut
- Dominos Pizza

Generic food prompts

- a cooked burger
- a serving of fries
- fried chicken pieces
- a slice of pizza
- a pizza box on the table
- fast food meal

Negative prompts

- phone home menu
- texting on a phone

- scrolling apps
- using calculator
- buying clothes online
- doing homework
- writing notes
- taking a quiz
- drawing or painting
- filling forms

A.2.2 Prompt-to-Brand Mapping Rules

Generic food prompts are mapped to candidate brands based on domain knowledge of fast-food product associations. This mapping is applied only when a corresponding visual detection is already present.

Table A.1: CLIP prompt groups and associated brand candidates

Prompt group	Candidate brands
Generic A (burger, fries, fried chicken)	McDonalds, Burger King, KFC
Generic B (pizza-related items)	Pizza Hut, Dominos Pizza
Generic C (general fast food)	All target brands

A.2.3 Semantic Reinforcement Logic

Only the top-1 CLIP prediction is considered. If the predicted label corresponds to a direct brand prompt, a strong or weak semantic confirmation is applied depending on the confidence score. If the predicted label corresponds to a generic food prompt, a weighted bonus is distributed to the associated candidate brands. Negative prompts do not contribute to any reinforcement.

```
If top-1 CLIP label is a brand prompt:
    If confidence >= strong threshold:
        Confirm brand detection
```

```
    Else if confidence >= weak threshold:
        Add weighted semantic bonus

If top-1 CLIP label is a generic food prompt:
    Distribute weighted bonus to candidate brands
```

A.3 Pseudo-code for Frame Extraction Procedure

```
Algorithm: Content-aware Frame Extraction

Input: Screen-recorded video V
(processed as a single segment in the current implementation)
Output: Set of extracted frames F

Initialise F <- empty
Initialise last_saved_frame <- None
Initialise last_saved_time <- None
Initialise last_diff_saved_time <- None

For each frame fi at time t in the current segment of V:
    If fi is the first frame:
        Add fi to F
        last_saved_frame <- fi
        last_saved_time <- t
    Else if t - last_saved_time >= 2 seconds:
        Add fi to F
        last_saved_frame <- fi
        last_saved_time <- t
    Else:
        Compute pixel difference d between fi and last_saved_frame
        If d >= 5000 and
           (last_diff_saved_time is None or t - last_diff_saved_time >= 1 second):
            Add fi to F
            last_saved_frame <- fi
            last_diff_saved_time <- t
```

```
            // note: last_saved_time is NOT updated for change-triggered frames

Ensure the final frame of each processed segment is retained
Resize all frames so that max edge = 800 pixels
Return F
```

A.4 Repository Structure and Output Definitions

This section documents the organisation of the project repository to support transparency and reproducibility. All datasets, intermediate outputs, trained models, and evaluation results are stored under a unified root directory:

```
Z:\2024\STUDY DATA\A_Semi_Automated_Pipeline_for_Fast_Food_Brand
_Exposure_Detection_Any
```

The repository follows a structured layout organised by processing stage. A template overview is shown below, where `<candidate_id>` and `<video_name>` denote placeholders repeated for each participant and recording.

```
A_Semi_Automated_Pipeline_for_Fast_Food_Brand_Exposure_Detection_Any/
- Extracted Frame-2023/
   - <candidate_id>/
      - <video_name>/
         - frames/
         - detail_<video_name>.csv
         - keep_ranges.csv
      - summary.csv
- Extracted Frame-2024/ (same structure as 2023)
- 1-YOLO-Training/
   - CleanLogo-Only/
      - Dataset/ (S/M/L zipped datasets)
      - Train-Result-<model>-<scale>/
         - weights/ (best.pt, last.pt)
         - args.yaml
         - results.csv, results.png
```

```
      - confusion_matrix.png, confusion_matrix_normalized.png
      - BoxF1_curve.png, BoxP_curve.png, BoxR_curve.png, BoxPR_curve.png
      - train_batch*.jpg
      - val_batch*_pred.jpg, val_batch*_labels.jpg
  - CleanLogo + VideoMix/ (same structure)
  - CleanLogo + VideoMix + HardNeg/ (same structure)
- 2-YOLO-Evaluation/
  - 1-instance-level evaluation/
    - DataSet/
    - TRAIN-RESULT/
    - Models Evaluation - eval#1.xlsx
  - 2-frame-level validation/
    - DataSet/
    - eval#2-Cal/
      - 1-conf-swept-hit-details/
      - ensemble-weight-sweep.py
  - 3-large-scale raw frame Cross-check/
    - eval#3_55k_frame-base_multiclass_cross-check-v2/
- 3-Pipeline/
  - eval#2_E03@0.5/                     # experiment-specific outputs
  - Figure 4.1 Pipeline.pdf
  - eval2-1k_Pipeline_summary.xlsx       # aggregated pipeline summary (eval2)
  - eval#3_55k_Pipeline_summary.xlsx     # aggregated pipeline summary (eval3)
  - yolo-ocr-clip-pieline-FIN.py
- 4-Application/
  - 2023-Newlands-Result/
    - TP/
    - FP/
    - NON-TARGET_HIT/
    - summary.xlsx
    - exposure_summary_by_brand.xlsx
  - 2024-Newlands-Result/ (same structure as 2023)
  - 0-Preprocessing-v9-no-black-screen-fillter.ipynb
  - apply-pipeline.py
  - exposure-cal.py
```

A.4.1 Key Output File Definitions

Several core CSV and model files are produced throughout the pipeline:

- **frames/**: Contains resized extracted frames (maximum edge 800 pixels) used for subsequent detection and evaluation.

- **Train-Result-* folders**: Contain standard YOLO training artefacts, including configuration arguments (`args.yaml`), final model weights (`best.pt`), epoch-level metrics (`results.csv`), confusion matrices, precision–recall and F1 curves, training batch visualisations, and validation prediction examples. These outputs support model selection and performance reporting.

- **Evaluation folders**: Instance-level evaluation (eval#1) is performed using standard YOLO validation outputs within the training folders. Frame-level evaluation (eval#2, eval#3) contain frame-level validation and large-scale cross-check results, including ensemble sweeps in CSV/XLSX format and performance figures (e.g., pass-confidence versus recall or false-positive rate), supporting threshold and weight selection.

- **4-Application**: Contains the final deployment stage of the pipeline on real-world Kids Online data. This directory includes preprocessing notebooks, executable scripts for running the integrated YOLO+OCR+CLIP pipeline, exposure aggregation utilities, and year-specific result folders.

All candidates and recordings follow the same directory conventions. Only representative examples are shown above for clarity.

References

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